Math 401 — Applications of Linear Algebra

Exercise Solutions (SP1–SP5)

Review · Least Squares · Curve Fitting · Team Ranking · Graph Theory · Markov Chains · PageRank · SVD · Image Compression · Quantum Qubit

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Chapter 1

Linear Algebra Review — Solutions

Source: Justin Wyss-Gallifent, Math 401/431: Linear Algebra Review. These exercises cover the prerequisite material pointed to in SP1.

Section 7: Eigenvalues and Eigenvectors

Exercise 7.1. Find the characteristic polynomials and eigenvalues of:
(a) $A = \begin{pmatrix} 4 & -5 \\ 1 & -2 \end{pmatrix}$ (b) $A = \begin{pmatrix} 2 & -1 \\ 1 & 4 \end{pmatrix}$ (c) $A = \begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix}$

Solution. **(a)** $\det(\lambda I - A) = (\lambda - 4)(\lambda + 2) + 5 = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1)$.

$$\boxed{\lambda_1 = 3, \quad \lambda_2 = -1.}$$

Eigenvector for $\lambda_1 = 3$: $(3I - A) = \begin{pmatrix} -1 & 5 \\ -1 & 5 \end{pmatrix} \to v_1 = \begin{pmatrix} 5 \\ 1 \end{pmatrix}$.

Eigenvector for $\lambda_2 = -1$: $(-I - A) = \begin{pmatrix} -5 & 5 \\ -1 & 1 \end{pmatrix} \to v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

(b) $\det(\lambda I - A) = (\lambda - 2)(\lambda - 4) + 1 = \lambda^2 - 6\lambda + 9 = (\lambda - 3)^2$.

$$\boxed{\lambda = 3 \text{ (algebraic multiplicity 2).}}$$

$(3I - A) = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} \to$ eigenspace spanned by $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ (geometric multiplicity 1; not diagonalizable over $\mathbb{R}$).

(c) $\det(\lambda I - A) = (\lambda + 1)^2 + 4 = \lambda^2 + 2\lambda + 5$.

$$\lambda = \frac{-2 \pm \sqrt{4 - 20}}{2} = -1 \pm 2i.$$

$$\boxed{\lambda_{1,2} = -1 \pm 2i.}$$

(No real eigenvalues; the matrix represents a rotation-plus-scaling in $\mathbb{R}^2$.) $\square$

Section 9: Linear Transformations

Exercise 9.1. Determine whether each transformation is linear. If not, give a counterexample. (a) $T(x_1, x_2) = (x_1 + 2x_2,\ x_1,\ -3x_2)$ (b) $T(x_1, x_2, x_3) = (-x_1,\ x_1 + x_2)$ (c) $T(x_1, x_2) = (x_1 + 2,\ 3x_2)$

Solution. **(a) Linear.**

$$T(\alpha u + \beta v) = \big(\alpha u_1 + \beta v_1 + 2(\alpha u_2 + \beta v_2),\ \alpha u_1 + \beta v_1,\ -3(\alpha u_2 + \beta v_2)\big) = \alpha T(u) + \beta T(v).$$

Representing matrix: $A = \begin{pmatrix} 1 & 2 \\ 1 & 0 \\ 0 & -3 \end{pmatrix}$.

(b) Linear.

$$T(\alpha u + \beta v) = \big(-(\alpha u_1 + \beta v_1),\ (\alpha u_1 + \beta v_1) + (\alpha u_2 + \beta v_2)\big) = \alpha T(u) + \beta T(v).$$

Representing matrix: $A = \begin{pmatrix} -1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}$.

(c) Not linear.

By Theorem 9.0.1, any linear transformation must satisfy $T(\mathbf{0}) = \mathbf{0}$.

$$T(0, 0) = (0 + 2,\ 3 \cdot 0) = (2, 0) \neq (0, 0).$$

So T is not linear. Explicit violation: $T(2 \cdot (1, 0)) = T(2, 0) = (4, 0)$ but $2T(1, 0) = 2(3, 0) = (6, 0) \neq (4, 0)$. □ □

Exercise 9.3. Use the matrix method (Theorem 9.0.2) to check if $T(x_1, x_2) = (5x_1 + 7x_2,\ 5x_1 - x_2)$ is linear.

Solution. **Step 1: Assume linearity and find candidate matrix.**

$$T(e_1) = T(1, 0) = (5, 5), \quad T(e_2) = T(0, 1) = (7, -1).$$

Candidate matrix: $M = \begin{pmatrix} 5 & 7 \\ 5 & -1 \end{pmatrix}$.

Step 2: Verify $T(x_1, x_2) = Mx$.

$$M \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 5x_1 + 7x_2 \\ 5x_1 - x_2 \end{pmatrix} = T(x_1, x_2). \quad \checkmark$$

Conclusion: T is linear. The representing matrix is $M = \begin{pmatrix} 5 & 7 \\ 5 & -1 \end{pmatrix}$. □

Exercise 9.4. Use the matrix method to check if $T(x_1, x_2) = (x_1^2 x_2,\ x_1 - x_2)$ is linear.

Solution. **Step 1: Assume linearity and find candidate matrix.**

$$T(e_1) = T(1, 0) = (1^2 \cdot 0,\ 1 - 0) = (0, 1), \quad T(e_2) = T(0, 1) = (0^2 \cdot 1,\ 0 - 1) = (0, -1).$$

Candidate matrix: $M = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}$.

Step 2: Verify $T(x_1, x_2) = Mx$.

$$M \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ x_1 - x_2 \end{pmatrix}.$$

But $T(x_1, x_2) = (x_1^2 x_2,\ x_1 - x_2)$. The first component is $x_1^2 x_2$, not 0.

Contradiction. We have a specific counterexample: let $x_1 = 2, x_2 = 1$:

$$T(2,1) = (4 \cdot 1,\ 2-1) = (4,1), \quad M\begin{pmatrix}2\\1\end{pmatrix} = (0,1) \neq (4,1).$$

Conclusion: T is not linear. $\square$

(Alternative direct proof: $T(2e_1) = T(2,0) = (0,2)$ but $2T(e_1) = 2(0,1) = (0,2)$ — this particular check passes. But $T(e_1+e_2) = T(1,1) = (1,0)$ while $T(e_1)+T(e_2) = (0,1)+(0,-1) = (0,0) \neq (1,0)$.) $\square$

Chapter 2

Least Squares — Solutions

Exercise 3.1. Find the least-squares solution and error for $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 0 & 2 \\ 5 & 1 & 1 \\ 2 & 2 & 0 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$.

Solution. Normal equations $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$:

$$A^T A = \begin{pmatrix} 31 & 11 & 6 \\ 11 & 9 & 10 \\ 6 & 10 & 14 \end{pmatrix}, \quad A^T\mathbf{b} = \begin{pmatrix} 24 \\ 13 \\ 10 \end{pmatrix}.$$

```
A = [1 2 3; -1 0 2; 5 1 1; 2 2 0]; b = [1;2;3;4];
x_hat = inv(A'*A)*A'*b;  ls_error = norm(A*x_hat-b);
```

Output: $\hat{\mathbf{x}} \approx (0.3619,\ 0.9281,\ 0.0951)^T$, error ≈ 3.0035. □

Exercise 3.2. Find the vector in $\mathrm{Col}(A)$ closest to $\mathbf{b}$ for $A = \begin{pmatrix} 1 & 2 \\ 0 & -3 \\ 2 & 6 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

Solution. $A^T A = \begin{pmatrix} 5 & 14 \\ 14 & 49 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $\det = 49$.

$$\hat{\mathbf{x}} = \frac{1}{49}\begin{pmatrix} 49 & -14 \\ -14 & 5 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 9/7 \\ -19/49 \end{pmatrix}.$$

Closest vector: $A\hat{\mathbf{x}} = \begin{pmatrix} 25/49 \\ 57/49 \\ 12/49 \end{pmatrix} \approx \begin{pmatrix} 0.510 \\ 1.163 \\ 0.245 \end{pmatrix}$. □

Exercise 3.3. Find the vector in $\mathrm{span}\left\{\begin{pmatrix} 1 \\ 2 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 3 \\ 0 \\ 3 \end{pmatrix}\right\}$ closest to $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$.

Solution. $A = \begin{pmatrix} 1 & 3 \\ 2 & 3 \\ -1 & 0 \\ 0 & 3 \end{pmatrix}$. Normal equations: $A^T A = \begin{pmatrix} 6 & 6 \\ 6 & 27 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$. $\det = 162 - 36 = 126$. $\hat{\mathbf{x}} = \begin{pmatrix} -1/3 \\ 2/9 \end{pmatrix}$. Closest vector: $A\hat{\mathbf{x}} = \begin{pmatrix} 1/3 \\ 0 \\ 1/3 \\ 2/3 \end{pmatrix}$. □

Exercise 3.4. Can $A^T A\mathbf{x} = A^T\mathbf{b}$ have no solutions?

Solution. **No.** The projection $\mathrm{Proj}_{\mathrm{Col}(A)}\,\mathbf{b}$ always exists, so the normal equation always has at least one solution. □

Exercise 3.5. Can more variables than equations give a unique least-squares solution?

Solution. **No.** If A is $m \times n$ with $n > m$, the columns of A are linearly dependent (at most m can be independent). Dependent columns $\Rightarrow$ infinitely many least-squares solutions. □

Exercise 3.6. Find the vector in $\mathrm{span}\left\{\begin{pmatrix}1\\2\end{pmatrix}\right\}$ closest to $\begin{pmatrix}5\\5\end{pmatrix}$.

Solution. $A = \begin{pmatrix}1\\2\end{pmatrix}$, $\mathbf{b} = \begin{pmatrix}5\\5\end{pmatrix}$. Normal equation: $5\hat{t} = 15 \Rightarrow \hat{t} = 3$. Closest vector: $\begin{pmatrix}3\\6\end{pmatrix}$. □

Exercise 3.7. Find the point on $y = 3x$ closest to $(2, 3)$.

Solution. Matrix form: $\begin{pmatrix}1\\3\end{pmatrix} t = \begin{pmatrix}2\\3\end{pmatrix}$. Normal equation: $10\hat{t} = 11 \Rightarrow \hat{t} = 1.1$. **Closest point:** $(1.1,\ 3.3)$. □

Exercise 3.8. Show the least-squares formula gives the exact solution when A is invertible.

Solution. $(A^T A)^{-1} A^T\mathbf{b} = A^{-1}(A^T)^{-1}A^T\mathbf{b} = A^{-1}\mathbf{b}$. This is the exact solution. □ □

Exercise 3.9. Show $x + 2y = 4$, $x + 2y = 3$ has no unique least-squares solution.

Solution. $A = \begin{pmatrix}1 & 2\\1 & 2\end{pmatrix}$ has dependent columns. $A^T A = \begin{pmatrix}2 & 4\\4 & 8\end{pmatrix}$ is singular. The augmented system row-reduces to $\left(\begin{array}{cc|c}2 & 4 & 7\\0 & 0 & 0\end{array}\right)$, giving infinitely many solutions $\hat{\mathbf{x}} = \begin{pmatrix}7/2 - 2t\\t\end{pmatrix}$. □

Exercise 3.10. $A = \begin{pmatrix}1 & 2\\-1 & 1\\0 & 3\end{pmatrix}$, $\mathbf{b} = \begin{pmatrix}1\\3\\0\end{pmatrix}$. (a) Solve via Gram-Schmidt; (b) via normal equations.

Solution. **(a) Gram-Schmidt:** $\mathbf{c}_1 = (1, -1, 0)^T$, $\mathbf{c}_2 = (2, 1, 3)^T$.

$$\mathbf{c}_2^{\perp} = (2, 1, 3)^T - \frac{1}{2}(1, -1, 0)^T = (3/2, 3/2, 3)^T.$$

$$\hat{\mathbf{b}} = \frac{-2}{2}(1, -1, 0)^T + \frac{3 + 9/2 + 0}{9/4 + 9/4 + 9}(3/2, 3/2, 3)^T \approx (-1/3, 5/3, 4/3)^T.$$

Solving $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$ gives $\hat{\mathbf{x}} \approx (-1.222,\ 0.444)^T$.

(b) Normal equations: $A^T A = \begin{pmatrix}2 & -3\\-3 & 14\end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix}-2\\5\end{pmatrix}$. $\det = 19$. $\hat{\mathbf{x}} = \frac{1}{19}\begin{pmatrix}14 & 3\\3 & 2\end{pmatrix}\begin{pmatrix}-2\\5\end{pmatrix} = \frac{1}{19}\begin{pmatrix}-13\\8\end{pmatrix} \approx \begin{pmatrix}-0.684\\0.421\end{pmatrix}$.

Hmm – let me re-check: $A^T A = \begin{pmatrix}1^2 + 1^2 & 1 \cdot 2 + (-1) \cdot 1\\2 \cdot 1 + 1 \cdot (-1) & 2^2 + 1^2 + 3^2\end{pmatrix} = \begin{pmatrix}2 & 1\\1 & 14\end{pmatrix}$. $A^T\mathbf{b} = \begin{pmatrix}1 - 3\\2 + 3\end{pmatrix} = \begin{pmatrix}-2\\5\end{pmatrix}$. $\det = 27$. $\hat{\mathbf{x}} = \frac{1}{27}\begin{pmatrix}14 & -1\\-1 & 2\end{pmatrix}\begin{pmatrix}-2\\5\end{pmatrix} = \frac{1}{27}\begin{pmatrix}-33\\12\end{pmatrix} \approx \begin{pmatrix}-1.222\\0.444\end{pmatrix}$. Both agree. ✓ □

Exercise 3.11. $A = \begin{pmatrix} 2 & -2 \\ 1 & 4 \\ 1 & 2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}$.

Solution. $A^T A = \begin{pmatrix} 6 & 2 \\ 2 & 24 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 2 \\ 16 \end{pmatrix}$. $\det = 140$. $\hat{\mathbf{x}} = \frac{1}{140}\begin{pmatrix} 24 & -2 \\ -2 & 6 \end{pmatrix}\begin{pmatrix} 2 \\ 16 \end{pmatrix} = \frac{1}{140}\begin{pmatrix} 16 \\ 92 \end{pmatrix} \approx \begin{pmatrix} 0.114 \\ 0.657 \end{pmatrix}$.

Error: $\|A\hat{\mathbf{x}} - \mathbf{b}\| \approx 0.507$. □

Exercise 3.12. Explain why a least-squares problem always has a solution.

Solution. $\mathrm{Col}(A)$ is a subspace of $\mathbb{R}^m$. Every point in $\mathbb{R}^m$ has a unique orthogonal projection onto any subspace. So $\hat{\mathbf{b}} = \mathrm{Proj}_{\mathrm{Col}(A)}\,\mathbf{b}$ always exists, $\hat{\mathbf{b}} \in \mathrm{Col}(A)$, and $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$ is consistent. □

Exercise 3.13. What is wrong with $(A^T A)^{-1}A^T = A^{-1}(A^T)^{-1}A^T = A^{-1}$?

Solution. In the least-squares setting, A is typically $m \times n$ with $m > n$ (not square), so A^{-1} **does not exist**. Similarly $(A^T)^{-1}$ does not exist. The factorization $(BC)^{-1} = C^{-1}B^{-1}$ is only valid when both B and C are invertible square matrices. □

Exercise 3.14. Why does switching columns of A not affect the least-squares solution?

Solution. Switching columns multiplies A on the right by a permutation P: $A' = AP$. Since $\mathrm{Col}(A') = \mathrm{Col}(A)$, the projection $\mathrm{Proj}_{\mathrm{Col}(A')}\,\mathbf{b} = \mathrm{Proj}_{\mathrm{Col}(A)}\,\mathbf{b}$ is unchanged, and so is the minimum error. The solution changes as $\hat{\mathbf{x}}' = P^T\hat{\mathbf{x}}$ (reordered to match the new column order), but the residual $A'\hat{\mathbf{x}}' - \mathbf{b} = A\hat{\mathbf{x}} - \mathbf{b}$ is unchanged. □

Chapter 3

Curve Fitting — Solutions

Exercise 4.1. Points $(-1,0),(1,2),(2,2)$: (a) find LS line; (b) what is minimized?

Solution. $A = \begin{pmatrix} -1 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$. $A^TA = \begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$, det $= 14$. $\hat{\mathbf{c}} = \frac{1}{14}\begin{pmatrix} 3 & -2 \\ -2 & 6 \end{pmatrix}\begin{pmatrix} 5 \\ 4 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 11/7 \end{pmatrix}$.

LS line: $f(x) = \frac{1}{2}x + \frac{11}{7}$.

(b) The LS line minimizes $\sum(y_i - f(x_i))^2$, the sum of squared *vertical* distances. □

Exercise 4.2. Points $(1,1),(5,2),(6,2),(8,3)$: (a) LS line; (c) estimate y at $x = 20$.

Solution.

```
pts=[1 1;5 2;6 2;8 3]; A=[pts(:,1),ones(4,1)]; b=pts(:,2);
x_hat=inv(A'*A)*A'*b;
```

$m \approx 0.2692$, $b_0 \approx 0.6538$. **Line:** $f(x) \approx 0.269x + 0.654$.

(c): $f(20) \approx 0.269(20) + 0.654 \approx 6.04$. □

Exercise 4.3. Points $(-1,3),(0,1),(1,2),(3,9)$: (a) LS parabola; (c) solve $f(x) = 10$.

Solution. $A = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \\ 9 & 3 & 1 \end{pmatrix}$.

```
pts=[-1 3;0 1;1 2;3 9]; A=[pts(:,1).^2,pts(:,1),ones(4,1)]; b=pts(:,2);
x_hat=inv(A'*A)*A'*b;
```

$a \approx 1.045$, $b \approx -0.555$, $c \approx 1.291$. **Parabola:** $f(x) \approx 1.045x^2 - 0.555x + 1.291$.

(c): $1.045x^2 - 0.555x + 1.291 = 10 \Rightarrow 1.045x^2 - 0.555x - 8.709 = 0$. $x = \frac{0.555 \pm \sqrt{0.308 + 36.41}}{2.09} \Rightarrow$ $x \approx 3.16$ or $x \approx -2.63$. □

Exercise 4.4. Points $(-2,0),(0,3),(4,4)$: LS $f(x) = ae^x + b$; estimate $f'(1)$.

Solution. $A = \begin{pmatrix} e^{-2} & 1 \\ 1 & 1 \\ e^4 & 1 \end{pmatrix}$.

```
pts=[-2 0;0 3;4 4]; A=[exp(pts(:,1)),ones(3,1)]; b=pts(:,2);
x_hat=inv(A'*A)*A'*b;
```

$a \approx 0.04693$, $b \approx 1.4615$. **Function:** $f(x) \approx 0.04693e^x + 1.4615$.

$f'(1) = ae \approx 0.04693 \times 2.718 \approx 0.128$. □

Exercise 4.5. Points $(-2, 6.3), (3, 1.2), (5, 7.1), (8, -2.8), (9, -0.05)$: LS $f(x) = a + b \sin x$, estimate $f(\pi/2)$.

Solution.

```
pts=[-2 6.3;3 1.2;5 7.1;8 -2.8;9 -0.05];
A=[ones(5,1),sin(pts(:,1))]; b=pts(:,2);
x_hat=inv(A'*A)*A'*b;
```

$a \approx 2.026$, $b \approx -4.974$.

$f(\pi/2) = 2.026 - 4.974 \sin(\pi/2) = 2.026 - 4.974 \approx -2.948$. □

Exercise 4.6. Points $(-3, -2, 45), (2, -2, 30), (0, 1, 6), (-2, 3, 55), (6, 5, 230)$: LS $f(x, y) = ax^2 + by^2$; estimate $f(3, 5)$.

Solution.

```
pts=[-3 -2 45;2 -2 30;0 1 6;-2 3 55;6 5 230];
A=[pts(:,1).^2,pts(:,2).^2]; b=pts(:,3);
x_hat=inv(A'*A)*A'*b;
```

$a \approx 3.027$, $b \approx 4.818$.

$f(3, 5) \approx 3.027(9) + 4.818(25) \approx 27.24 + 120.45 \approx 147.7$. □

Exercise 4.7. Points $(0, 0), (0, 1), (1, 1)$: find LS line $y = mx + b$ **and** $x = ny + c$. Show they differ.

Solution. **Line** $y = mx + b$**:** $A_1 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$, $\mathbf{b}_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$. $A_1^T A_1 = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$, $A_1^T \mathbf{b}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

$\hat{m} = 1/2$, $\hat{b} = 1/2$. **Line 1:** $y = \frac{1}{2}x + \frac{1}{2}$.

Line $x = ny + c$**:** Swap roles. $\hat{n} = 1$, $\hat{c} = -1/3$. **Line 2:** $x = y - \frac{1}{3}$, i.e., $y = x + \frac{1}{3}$ (slope 1).

Line 1 has slope 1/2; line 2 has slope 1. **Different**, because they minimize different distances (vertical vs. horizontal). □

Exercise 4.8. Orbital data (θ°, r): $(23, 152), (50, 135), (100, 102), (110, 110), (152, 137)$. Fit $Ar^2 \cos^2\theta + Br^2 \sin^2\theta = 1$.

Solution.

```
data=[23 152;50 135;100 102;110 110;152 137];
theta=data(:,1)*pi/180; r=data(:,2);
A=[(r.*cos(theta)).^2,(r.*sin(theta)).^2];
b=ones(5,1); x_hat=inv(A'*A)*A'*b;
```

$A \approx 3.976 \times 10^{-5}$, $B \approx 7.215 \times 10^{-5}$.

(b): At $\theta = 225^\circ$: both $\cos^2, \sin^2 = 1/2$, so $r^2 = 2/(A + B) \approx r \approx 136.5$ Mmiles.

(c): r is maximized when $A \cos^2\theta + B \sin^2\theta$ is minimized. Since $A < B$, minimum at $\theta = 0$: $r_{\max} = 1/\sqrt{A} \approx 158.6$ Mmiles. □

Exercise 4.10. Points $(1, 1), (2, 1), (3, 2), \ldots, (3, 2)$ (n times). Find $\lim_{n\to\infty} \hat{\mathbf{c}}_n$.

Solution. With $n + 2$ points: $A^T A = \begin{pmatrix} 9n + 5 & 3n + 3 \\ 3n + 3 & n + 2 \end{pmatrix}$, $A^T \mathbf{b} = \begin{pmatrix} 6n + 3 \\ 2n + 2 \end{pmatrix}$.

$\det = 5n + 1$. Computing: $m_n = \dfrac{3n}{5n + 1} \to \dfrac{3}{5}$, $b_n = \dfrac{n + 1}{5n + 1} \to \dfrac{1}{5}$.

$\lim_{n\to\infty} \hat{\mathbf{c}}_n = \begin{pmatrix} 3/5 \\ 1/5 \end{pmatrix}$.

Check: $y(3) = \frac{9}{5} + \frac{1}{5} = 2$ ✓ (line passes through $(3, 2)$ in the limit, as expected from infinite weight). □

Exercise 4.13. Which function templates allow LS?

Solution. LS requires all unknowns to appear *linearly*.

(a) $ax^2 + bx^c$: **No** (c is a nonlinear unknown exponent).

(b) $ae^x + bx$: **Yes** (linear in a, b; matrix column i: (e^{x_i}, x_i)).

(c) $e^{ax} + bx$: **No** (a appears in the exponent).

(d) $a \sin x + b \cos x + c$: **Yes** (linear in a, b, c).

(e) $a \sin(bx) + c$: **No** (b appears inside sine).

□

Chapter 4

Team Ranking — Solutions

Exercise 5.1. Four teams with given results. (a) Find rankings. (b) Predict T1 vs T4.

Solution. Games: T1–T2(1), T1–T3(1), T2–T3(2), T2–T4(1), T3–T4(1). Point totals: $q_1 = -1$, $q_2 = 13$, $q_3 = -8$, $q_4 = -4$ (check: sum $= 0$ ✓).

$$M = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 4 & -2 & -1 \\ -1 & -2 & 4 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad \bar{q} = \begin{pmatrix} -1 \\ 13 \\ -8 \\ 0 \end{pmatrix}.$$

Rankings: $r_1 = 0.125$, $r_2 = 2.375$, $r_3 = -1.125$, $r_4 = -1.375$. Order: T2>T1>T3>T4.

(b): Expected margin $= r_1 - r_4 = 1.5$. T1 expected to win by 1.5. □

Exercise 5.2. Four teams. (a) Find rankings. (b) Predict T2 vs T4.

Solution. Same M as 5.1 (same game counts). $\bar{q} = (38, -21, -23, 0)^T$. **Rankings:** $r_1 = 13.5$, $r_2 = -5.333$, $r_3 = -5.667$, $r_4 = -2.5$. Order: T1>T4>T2≈T3.

(b): $r_2 - r_4 = -2.833$. T4 expected to win by about 2.8. □

Exercise 5.3. Read off structural info from a given Massey equation.

Solution. $M = \begin{pmatrix} 3 & -1 & -2 & 0 \\ -1 & 4 & -2 & -1 \\ -2 & -2 & 4 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$.

(a) 4 teams. **(b)** T4 original diagonal entry: $|m_{42}| = 1$, so 1 game (vs T2). **(c)** T1: $m_{11} = 3$ games; $|m_{12}| = 1$ vs T2, $|m_{13}| = 2$ vs T3 (twice). **(d)** Original $m_{44} = 1$. Total $= (3 + 4 + 4 + 1)/2 = 6$ games. □

Exercise 5.4. Show the sum of diagonal entries of M (before modification) is always even.

Solution. $\sum_i m_{ii} = \sum_i (\text{games team } i \text{ played}) = 2G$ (each game is counted in exactly 2 teams' totals). Since $2G$ is even, done. □ □

Exercise 5.5. Four teams with home advantage. (a) Without home, (b) reduce home wins by 25%.

Solution. **(a):** $q_1 = 15, q_2 = -22, q_3 = 14, q_4 = -7$. Rankings: $r_1 = 6.5, r_2 = -4.0, r_3 = 2.0, r_4 = -4.5$.

(b): Reduce home wins: T1 at home beats T2 by $5 \to 3.75$; T3 at home beats T4 by $21 \to 15.75$. $q_1 = 3.75 + 10 = 13.75$, $q_2 = -3.75 - 10 + 7 - 14 = -20.75$, $q_3 = -10 - 7 + 15.75 = -1.25$,

$q_4 = 14 - 15.75 = -1.75$. Rankings (b): $r_1 \approx 4.125, r_2 \approx -4.375, r_3 \approx -1.125, r_4 \approx 1.375$. T4 moved above T3. □

Exercise 5.6. If T1 beats T2 by k, T2 beats T3 by k, T3 beats T1 by k: show equal rankings.

Solution. $q_1 = k - k = 0$, $q_2 = -k + k = 0$, $q_3 = -k + k = 0$. With $\bar{q} = \mathbf{0}$ and the constraint row, the unique solution is $\hat{\mathbf{r}} = \mathbf{0}$. All teams have equal ranking 0. □ □

Exercise 5.9 / 5.10. Better to beat team twice by 1 or once by 2?

Solution. **(a) Once by 2:** T1-T2(1), T2-T3(1), T1-T3(1) with margins 3, 2, 2. $M = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix}$, $\bar{q} = (5, -1, 0)^T$. Rankings: $r_1 \approx 1.667$.

(b) Twice by 1: T1-T3 now twice (total margin 2). $M = \begin{pmatrix} 3 & -1 & -2 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix}$, $\bar{q} = (5, -1, 0)^T$. Rankings: $r_1 \approx 1.067$.

One win by 2 gives T1 a higher ranking than two wins by 1 each. □

Exercise 5.11. (a) T1 beats T2 by n, T2 beats T3 by 1, T3 beats T1 by 1. Find $\lim_{n\to\infty} \hat{\mathbf{r}}_n$. (b) T1 beats T2 by 1 n times instead.

Solution. **(a):** $q_1 = n - 1$, $q_2 = -(n-1)$, $q_3 = 0$. By symmetry $r_3 = 0$, $r_1 = -r_2$. Row 1: $3r_1 = n - 1$, so $r_1 = (n-1)/3 \to \infty$.

(b): Now $m_{11} = n + 1$, $m_{12} = m_{21} = -n$. Same point totals but $(2n+1)r_1 = n - 1$, so $r_1 = (n-1)/(2n+1) \to 1/2$.

Winning once by n is unlimited; winning n times by 1 converges to a bounded limit. □

Exercise 5.12. T1 beats T2 by α, T2 beats T3 by β, T3 beats T1 by γ. When are all rankings equal?

Solution. $q_1 = \alpha - \gamma$, $q_2 = -\alpha + \beta$, $q_3 = -\beta + \gamma$. All zero $\Leftrightarrow \alpha = \beta = \gamma$.

Two teams can tie (e.g., $\alpha = 2, \beta = 4, \gamma = 0$ gives $r_1 = r_3$). Spread is unbounded as $\alpha \to \infty$. □

Exercise 5.13. Two groups $\{T1, T2, T3\}$ and $\{T4, T5, T6\}$ with no inter-group games. (a) Massey fails; (b) rank groups; (c) after T3 beats T4 by 1.

Solution. **(a):** Two disconnected components $\Rightarrow$ block-singular Massey matrix.

(b): Each group has $M_{\text{sub}} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix}$, $\bar{q} = (2, 1, 0)^T$. Rankings: $r_1 \approx 2.333, r_2 \approx 0.333, r_3 \approx -2.667$ (same for T4,T5,T6).

```
M6=[1 -1 0 0 0 0;-1 2 -1 0 0 0;0 -1 2 -1 0 0;
    0 0 -1 2 -1 0;0 0 0 -1 2 -1;1 1 1 1 1 1];
q6=[2;-2+3;-3+1;-1+2;-2+3;0]; r6=M6\q6;
```

(c) Output: $r_1 \approx 5.33, r_2 \approx 3.33, r_3 \approx 0.33, r_4 \approx -0.67, r_5 \approx -2.67, r_6 \approx -5.67$. □

Exercise 5.16. Nontrivial examples where no LS is needed; is sum constraint still required?

Solution. **Example:** T1 beats T2 by 3, T2 beats T3 by 3, T1 beats T3 by 6. The three equations $r_1 - r_2 = 3$, $r_2 - r_3 = 3$, $r_1 - r_3 = 6$ are consistent, so the original $A\mathbf{r} = \mathbf{p}$ has an exact solution.

Sum constraint still needed: Even with consistent results, the Massey matrix $A^T A$ is singular (its rows sum to 0). Without $\sum r_i = 0$, there are infinitely many solutions differing by a constant shift. The constraint gives the unique centered solution. $\square$

Chapter 5

Graph Theory — Solutions

Exercise 12.1. Graph: 1–2, 1–3, 2–3, 3–4. Count walks and explain.

Solution.

```
A=make_adj([1 2;1 3;2 3;3 4],4); fprintf('%d\n',A^3(1,3));
```

(a) $(A^3)_{13} = 4$ walks of length 3 from v1 to v3.

(b) $(A^{20})_{24} = 790748$.

(c) $(A^3)_{44} = 0$. Vertex 4 has degree 1 (only neighbor: v3). A closed walk v4→v3→?→v4 of length 3 would need to return from a neighbor of v3 to v4 in one step; but neighbors of v3 in $\{v1, v2, v4\}$ can only reach v4 via v3, which already used one step. No such walk exists. □

Exercise 12.2. Cycle C_5: 1–2–3–4–5–1. Count walks and identify patterns.

Solution. $(A^3)_{12} = 3$. $(A^{10})_{24} = 220$.

(c) Even walks $v3 \to v5$: $k = 2 : 1,\ k = 4 : 4,\ k = 6 : 15,\ k = 8 : 57,\ k = 10 : 220$. Growing because $v3$ and $v5$ are 2 apart and more paths accumulate with length.

(d) Odd walks $v2 \to v4$: $k = 1 : 0,\ k = 3 : 1,\ k = 5 : 5,\ k = 7 : 22,\ k = 9 : 93$. Growing similarly (v2 and v4 are 2 apart going one way, 3 the other). □

Exercise 12.3. Why does $\frac{1}{6}\operatorname{tr}(A^3)$ count triangles but $c \cdot \operatorname{tr}(A^4)$ doesn't count squares?

Solution. Every closed walk of length 3 in a simple graph is a triangle (no shortcuts possible). So $\operatorname{tr}(A^3)$ counts triangle traversals exactly.

Closed walks of length 4 include squares *and* backtracks ($v \to u \to v \to u \to v$) and triangle-with-backtrack paths. $\operatorname{tr}(A^4)$ therefore overcounts. The same issue applies to all cycle lengths ≥ 4. □

Exercise 12.4. Apply Fiedler to graph 1–2, 1–3, 2–3, 3–4.

Solution. $L = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$.

```
A=make_adj([1 2;1 3;2 3;3 4],4); L=make_lap(A);
[fv,lam2]=fiedler_vec(L);
```

Eigenvalues: $0, 1, 3, 4$. Fiedler value $\lambda_2 = 1$. Fiedler vector $\approx (-0.408, -0.408, 0, 0.816)^T$.

Partition: $V_- = \{1, 2\}$, $V_0 = \{3\}$ (bridge), $V_+ = \{4\}$. Cut = 1 edge (3–4). □

Exercise 12.5. Apply Fiedler to graph with two triangles sharing vertex 3.

Solution. Graph: 1–2, 1–3, 2–3, 3–4, 3–5, 4–5.

Fiedler vector $\approx (-0.5, -0.5, 0, 0.5, 0.5)^T$. Partition: $V_- = \{1, 2\}$, $V_+ = \{4, 5\}$. Vertex 3 is the bridge (can go in either group). □

Exercise 12.6. Fiedler vector $(-0.195, -0.632, -0.195, 0.512, 0.512)^T$: partition.

Solution. $V_- = \{1, 2, 3\}$ (negative entries), $V_+ = \{4, 5\}$ (positive entries). Partition: $(\{1, 2, 3\}, \{4, 5\})$. □

Exercise 12.7. (Requires the specific graph from the notes – Fiedler computation.)

Solution.

```
% Replace edge list below with the specific graph from Exercise 12.7
A=make_adj([...],n); L=make_lap(A); [fv,lam2]=fiedler_vec(L);
neg=find(fv<0); pos=find(fv>0);
```

Apply algorithm: build L, find Fiedler vector, partition by sign. □

Exercise 12.8. Match graphs G1, G2, G3 to Fiedler vectors $\bar{v}, \bar{w}, \bar{x}$.

Solution. G2 (linear chain): Fiedler vector is monotone → matches $\bar{v} = (-0.56, -0.41, -0.15, 0.15, 0.41, 0.56)$.

G1 (symmetric 2-wing): Two equal clusters → matches $\bar{w} = (0.46, 0.46, 0.26, -0.26, -0.46, -0.46)$.

G3 (asymmetric): matches $\bar{x} = (0.32, 0.51, 0.32, -0.12, -0.51, -0.51)$. □

Exercise 12.9. 8-vertex graph (from notes). (a) Apply Fiedler Method. (b) Draw components with dashed cross-edges. (c) Structural insights.

Solution. See the detailed solution in the Sample Exams document (Section 4, Exercise 12.9). In summary:

Build the Laplacian from the adjacency matrix given in the graph picture, then:

```
A=make_adj(edges,8); L=make_lap(A); [fv,lam2]=fiedler_vec(L);
neg=find(fv<0); pos=find(fv>0);
fprintf('V-: '); disp(neg'); fprintf('V+: '); disp(pos');
```

Partition by sign of the Fiedler vector entries. Draw the two components; cross-edges (edges between V_- and V_+) are drawn dashed, revealing the cut structure. □

Exercise 12.10. 8-vertex graph (second example, from notes). (a) Apply Fiedler Method. (b) Draw components with dashed cross-edges. (c) Structural insights.

Solution. Same procedure as Exercise 12.9. Construct the Laplacian for the graph pictured in the notes:

```
A=make_adj(edges,8); L=make_lap(A); [fv,lam2]=fiedler_vec(L);
neg=find(fv<0); pos=find(fv>0);
```

Partition V_- (negative Fiedler entries) vs. V_+ (positive entries). Draw each component separately; dashed edges mark the cut. The structural insight will depend on the specific graph: look for which cluster is more tightly connected (smaller internal Fiedler spread) and which cross-edges form the bottleneck. □

Exercise 12.11. 5-vertex graph. (a) Write down the Laplacian. (b) The Fiedler eigenspace is 2-dimensional; analyze all possible partitions methodically.

Solution. **(a)** Write the Laplacian $L = D - A$ from the adjacency structure given in the picture. For a 5-vertex graph with a 2-dimensional Fiedler eigenspace, the graph has a symmetry that causes $\lambda_2 = \lambda_3$.

(b) Let $\{v_1, v_2\}$ be a basis for the 2D eigenspace. Any unit vector $u = \cos\theta\, v_1 + \sin\theta\, v_2$ is a valid Fiedler vector. As θ varies from 0 to 2π, the sign pattern of u cycles through all possible partitions.

Methodical approach:

1. Find two orthonormal basis vectors v_1, v_2 of the eigenspace.
2. List all distinct sign-patterns of $\cos\theta\, v_1 + \sin\theta\, v_2$ by finding the angles θ where an entry changes sign (i.e., where $\cos\theta\,(v_1)_i + \sin\theta\,(v_2)_i = 0$, giving $\tan\theta = -(v_1)_i/(v_2)_i$).
3. For each interval between consecutive critical angles, record the sign pattern and the resulting partition.

```
A=make_adj(edges,5); L=make_lap(A);
[P,D]=eig(L); lams=diag(D);
idx=find(abs(lams-lams(2))<1e-8); % indices of Fiedler eigenvalue
v1=P(:,idx(1)); v2=P(:,idx(2));
thetas=atan2(-(v1),(v2)); % critical angles for each entry
thetas=sort(unique(mod(thetas,pi)));
for t=[thetas; thetas+pi/8]'
  u=cos(t)*v1+sin(t)*v2; disp(sign(u)');
end
```

□

Exercise 12.12. 6-vertex graph with 2D Fiedler eigenspace. (a) Find basis $\{v_1, v_2\}$. (b) Find multiple partitions. (c) Prove vertices 1&4, 2&5, 3&6 are always in different subgraphs.

Solution. **(a)** From the graph (two triangles connected symmetrically, e.g. the prism graph):

```
A=make_adj(edges,6); L=make_lap(A);
[P,D]=eig(L); lams=diag(D);
idx=find(abs(lams-lams(2))<1e-8);
v1=P(:,idx(1)); v2=P(:,idx(2));
fprintf('v1='); disp(v1'); fprintf('v2='); disp(v2');
```

(b) Try $u(\theta) = \cos\theta\, v_1 + \sin\theta\, v_2$ for several values of θ; each produces a different sign-partition. At least two distinct partitions will emerge (corresponding to the two "natural" 3-vs-3 splits).

(c) The graph has the symmetry $\sigma : i \mapsto i + 3 \pmod 6$ (swapping the two triangles). This symmetry sends each eigenvector v to $-v$ (the anti-symmetric mode). Therefore for any Fiedler vector $u = \cos\theta\, v_1 + \sin\theta\, v_2$:

$$u_{i+3} = -u_i \quad \text{for } i = 1, 2, 3.$$

Since u_i and u_{i+3} have opposite signs (and neither is 0 generically), vertices i and $i + 3$ always fall in opposite subgraphs. □ □

Exercise 12.13. K_n: what are the eigenvalues of the Laplacian?

Solution. $L_{K_n} = nI - J$ where $J = \mathbf{1}\mathbf{1}^T$. Eigenvectors of J: $\mathbf{1}$ (eigenvalue n) and any vector orthogonal to $\mathbf{1}$ (eigenvalue 0). So eigenvalues of $nI - J$: $n - n = 0$ (once) and $n - 0 = n$ ($n - 1$ times).

Guess: $\lambda(L_{K_n}) = \{0^{(1)}, n^{(n-1)}\}$.

Verification:

```
for n=[3 4 5 6], A=ones(n,n)-eye(n); L=diag(sum(A))-A;
fprintf('K_%d: ',n); fprintf('%.1f ',sort(eig(L))'); fprintf('\n'); end
```

□

Exercise 12.14. 11-vertex graph (from notes). (a) Find Fiedler vector. (b) Group sorted values into 3 subsets. (c) Partition into 3 subgraphs. (d) Draw. (e) Structural insights.

Solution. **(a)**

```
A=make_adj(edges,11); L=make_lap(A);
[fv,lam2]=fiedler_vec(L);
[fv_s,idx]=sort(fv);
fprintf('Sorted Fiedler entries:\n');
for i=1:11, fprintf('  v%d: %.4f\n',idx(i),fv_s(i)); end
```

(b) The sorted Fiedler values split into 3 groups separated by gaps:

- **Group A** (most negative): vertices with $v_i \ll 0$
- **Group B** (near zero): vertices with $v_i \approx 0$ (bridge/central)
- **Group C** (most positive): vertices with $v_i \gg 0$

Identify the two largest gaps in the sorted sequence to determine group boundaries.

(c) Assign each vertex to the group from (b). The three-way partition minimizes the number of cross-edges.

(d) Draw each group as a subgraph; draw dashed edges for cross-group connections.

(e) The three groups reveal which subsets of the 11-vertex graph are most internally cohesive, and the dashed cross-edges show the bottleneck connections. □

Exercise 12.15. Path graph $1 - 2 - \cdots - n$. (a) Predict Fiedler Method outcome before calculating; justify. (b) Check hypothesis for several values of n.

Solution. **(a) Prediction.**

The path P_n is a chain: vertices $1, 2, \ldots, n$ connected as i–$(i+1)$. There is no cycle; the graph is "one-dimensional." The natural partition into two connected halves is $\{1, \ldots, \lfloor n/2 \rfloor\}$ and $\{\lceil n/2 \rceil + 1, \ldots, n\}$, cutting exactly one edge.

Hypothesis: The Fiedler vector of P_n has entries monotonically increasing (or decreasing) from vertex 1 to n, with the cut at the midpoint (one cut-edge).

Justification: The Laplacian of P_n is the tridiagonal matrix $L_{ij} = -1$ for $|i - j| = 1$, $L_{ii} = 2$ for $1 < i < n$, and $L_{11} = L_{nn} = 1$. Its eigenvectors are known analytically: the k-th eigenvector has entries $v_i^{(k)} = \cos\left(\frac{(i-\frac{1}{2})(k-1)\pi}{n}\right)$. The Fiedler vector ($k = 2$) is $v_i \propto \cos\left(\frac{(i-\frac{1}{2})\pi}{n}\right)$, which is strictly monotone in i. The sign change occurs near $i = n/2$.

(b) Verification.

```
for n=[4 5 6 7 8]
  edges=[(1:n-1)',(2:n)'];
  A=make_adj(edges,n); L=make_lap(A);
  [fv,lam2]=fiedler_vec(L);
  fprintf('n=%d, lam2=%.4f, Fiedler=',n,lam2);
  fprintf('%.3f ',fv'); fprintf('\n');
end
```

For $n = 4$: $\lambda_2 = 2 - \sqrt{2} \approx 0.586$, $v \approx (-0.65, -0.27, 0.27, 0.65)$. Sign change between positions 2 and 3 $\Rightarrow$ partition $\{1, 2\} \mid \{3, 4\}$, cut = edge 2–3. ✓

For odd n: the middle vertex has $v_{(n+1)/2} \approx 0$ and can go in either group; the cut is still 1 edge. □

Exercise 12.16. 14-vertex graph (from notes). (a) Find Fiedler vector. (b) Separate into 3 groups. (c) Partition. (d) Draw. (e) Which vertex is most critical? (f) Structural insights.

Solution.

```
A=make_adj(edges,14); L=make_lap(A);
[fv,lam2]=fiedler_vec(L);
[fv_s,ord]=sort(fv);
fprintf('Sorted: '); fprintf('v%d:%.3f ',ord,fv_s); fprintf('\n');
```

(b) Inspect the sorted values for two large gaps. Label the three resulting groups A (negative), B (near-zero bridge), C (positive).

(c) The partition is $G_A \mid G_B \mid G_C$. Edges between groups are the cut-edges.

(d) Draw each subgraph separately; dashed lines show inter-group edges.

(e) The *most critical* vertex is the one in group B (bridge) with the Fiedler entry closest to 0 *and* the highest degree connecting to both other groups. Removing it would maximally disconnect the graph. If group B has only one vertex, that vertex is the critical articulation point.

(f) The three-group partition reveals hub-and-spoke structure or community structure. Look at whether the bridge group (B) is a single articulation point or a small cluster. □

Exercise 12.17. 13-vertex social network. (a) Use Fiedler Method to identify group most strongly connected to Person 10. (b) Why is the method ineffective?

Solution.

```
A=make_adj(edges,13); L=make_lap(A);
[fv,lam2]=fiedler_vec(L);
sign10=sign(fv(10));
group_with_10=find(sign(fv)==sign10);
fprintf('Group containing P10: '); disp(group_with_10');
```

(a) The Fiedler partition assigns Person 10 to the group with the same sign Fiedler entry. This group is the one identified as "most strongly connected" to P10 globally.

(b) Why ineffective: The Fiedler method is a *global* spectral method – it minimizes the total cut and finds a balanced partition of the entire graph. It does not find the local neighborhood of a specific vertex. Person 10 may have close friends (short graph-distance neighbors) who end up in the *opposite* Fiedler partition if they happen to be on the "wrong" side of the global minimum-cut. A better approach for finding P10's strongest connections is BFS/DFS, clustering coefficients, or ego-network analysis, not the Fiedler partition. □ □

SP3 Quiz: Graph Theory Problems

Exercise SP3 Quiz 1. Give examples of graphs Γ_1 and Γ_2 with the *same number of triangles* but which are *not isomorphic*. Fully justify using a theorem or definition.

Solution. **Example.**

- Γ_1: Two disjoint triangles, $K_3 \sqcup K_3$ (6 vertices, 6 edges, **2 triangles**, not connected).
- Γ_2: Triangular prism graph (6 vertices, 9 edges, **2 triangles**, connected, 3-regular). Its edges are $\{1\text{-}2, 2\text{-}3, 1\text{-}3, 4\text{-}5, 5\text{-}6, 4\text{-}6, 1\text{-}4, 2\text{-}5, 3\text{-}6\}$.

Same triangle count: Γ_1 has triangles $\{1, 2, 3\}$ and $\{4, 5, 6\}$ (2 total). Γ_2 has triangles $\{1, 2, 3\}$ and $\{4, 5, 6\}$ (2 total). ✓

Not isomorphic: By Definition 12.2.0.3, Γ_1 is *not connected* (it has two separate pieces). Γ_2 *is connected*. Connectivity is preserved under isomorphism (an isomorphism is a bijection on vertices preserving adjacency, so paths are preserved). Therefore $\Gamma_1 \ncong \Gamma_2$. □ □

Exercise SP3 Quiz 2. Give examples of: (a) a graph that is not simple but has a simple subgraph; (b) a graph that is not connected but has a connected subgraph.

Solution. **(a) Non-simple graph with a simple subgraph.**

Example: Let G have vertices $\{1, 2\}$ and two parallel edges between them (a multigraph, violating "no multiple edges" in Definition 12.2.0.2, hence not simple). The subgraph retaining only *one* of the two edges is K_2 (the complete graph on 2 vertices), which is simple.

Note: Any non-simple graph obtained by adding loops or multi-edges to a simple graph trivially contains the original simple graph as a subgraph.

(b) Non-connected graph with a connected subgraph.

Example: $G = K_3 \sqcup K_2$ (a triangle on vertices $\{1, 2, 3\}$ and a separate edge on vertices $\{4, 5\}$). G is not connected (vertices 1 and 4 are in different components). The subgraph K_3 (induced on $\{1, 2, 3\}$) is connected. □ □

SP3 Exam: Additional Graph Theory Problems

Exercise SP3 Exam: Graph Isomorphism. (a) How many 3×3 permutation matrices are there? (b) Prove $K_3 \ncong P_3$ by checking all permutation matrices. (c) Use the triangle theorem.

Solution. **(a)** $3! = 6$.

(b) $A_1 = K_3$ (triangle), $A_2 = P_3$ (path). None of the 6 permutations gives $PA_2P^T = A_1$.

```
A1=[0 1 1;1 0 1;1 1 0]; A2=[0 1 0;1 0 1;0 1 0];
found=false;
for k=1:6, P=eye(3)(perms([1 2 3])(k,:),:);
  if all(all(P*A2*P'==A1)), found=true; disp('equal'); end; end
if ~found, disp('Not isomorphic: no permutation works.'); end
```

Output: `Not isomorphic: no permutation works.`

(c) By Theorem 12.3.0.2 (triangles $= \frac{1}{6}\operatorname{tr}(A^3)$): K_3 has $\operatorname{tr}(A_1^3)/6 = 6/6 = 1$ triangle; P_3 has $\operatorname{tr}(A_2^3)/6 = 0/6 = 0$ triangles. Since isomorphic graphs have the same number of triangles, $K_3 \ncong P_3$. □ □

Exercise SP3 Exam: Planar and Regular Subgraphs. (a) Is it possible to find a *planar* graph with a *nonplanar* subgraph?
(b) Is it possible to find a *regular* graph with a *nonregular* subgraph?
Give an example or explain why none exist.

Solution. **(a) No, a planar graph cannot have a nonplanar subgraph.**

Proof: By Kuratowski's theorem, a graph is *nonplanar* if and only if it contains a subdivision of K_5 or $K_{3,3}$ as a subgraph. If H is a subgraph of G and H is nonplanar, then H contains a subdivision of K_5 or $K_{3,3}$, which is also a subgraph of G. But then G is nonplanar – contradiction. □

Intuition: Planarity is a "downward closed" property: removing vertices or edges from a planar graph always leaves a planar graph.

(b) Yes, a regular graph can have a nonregular subgraph.

Example: $G = K_4$ (complete graph on 4 vertices) is 3-regular. Consider the subgraph H obtained by removing the single edge $\{1, 2\}$:

$$\deg_H(1) = 2, \quad \deg_H(2) = 2, \quad \deg_H(3) = 3, \quad \deg_H(4) = 3.$$

H has vertices of degree 2 and degree 3, so H is not regular.

H is a valid subgraph of K_4 (retaining all 4 vertices, removing one edge). □

Note: Regularity is *not* a hereditary (subgraph-monotone) property, unlike planarity. □

True/False (from Sol. 12):

"Same adjacency matrix $\Rightarrow$ isomorphic": **True**. Identical adjacency matrices (with the same vertex labeling) means the graphs have exactly the same edge structure, hence $P = I$ witnesses isomorphism. $\square$

"Same degree matrix $\Rightarrow$ isomorphic": **False**. Counterexample: C_6 (6-cycle, every vertex degree 2) and $2K_3$ (two disjoint triangles, every vertex degree 2) both have degree sequence $(2, 2, 2, 2, 2, 2)$ but are not isomorphic (C_6 is connected, $2K_3$ is not).

Chapter 6

Markov Chains — Solutions

Exercise 6.1. Population diagram (3 states): (a) transition matrix; (b) asymmetry; (c) 1 and 2 iterations; (d) steady state; (e) smallest k for 4-decimal accuracy.

Solution. $T = \begin{pmatrix} 0.80 & 0.05 & 0.25 \\ 0.12 & 0.90 & 0.05 \\ 0.08 & 0.05 & 0.70 \end{pmatrix}$, $\mathbf{x}_0 = (0.6, 0.2, 0.2)^T$.

(b) T not symmetric $\Rightarrow$ movement rates differ by direction.

(c) $\mathbf{x}_1 = T\mathbf{x}_0 = (0.540, 0.262, 0.198)^T$; $\mathbf{x}_2 = T^2\mathbf{x}_0 \approx (0.497, 0.303, 0.200)^T$.

(d) Solve $(T - I)\mathbf{x}^* = \mathbf{0}$ with $\sum x_i = 1$: $\mathbf{x}^* \approx (0.416, 0.416, 0.168)^T$.

(e)

```
T=[0.80 0.05 0.25;0.12 0.90 0.05;0.08 0.05 0.70];
x=T^1*[0.6;0.2;0.2]; xs=steady_state(T); k=1;
while max(abs(x-xs))>5e-5, k=k+1; x=T*x; end; fprintf('k=%d\n',k);
```

□

Exercise 6.2. 6×6 transition matrix: (a) diagram; (b) regular?

Solution. $T = \begin{pmatrix} 0.9 & 0 & 0.05 & 0 & 0.15 & 0.05 \\ 0 & 0.8 & 0.05 & 0.2 & 0.05 & 0.05 \\ 0.01 & 0.1 & 0.5 & 0 & 0 & 0.05 \\ 0.02 & 0.1 & 0.1 & 0.75 & 0 & 0.05 \\ 0.07 & 0 & 0.1 & 0 & 0.7 & 0.05 \\ 0 & 0 & 0.2 & 0.05 & 0.1 & 0.75 \end{pmatrix}$.

(b) Regular, since T^2 has all positive entries. All states communicate (every state can reach every other state in at most 2 steps). □

Exercise 6.3. Construct a 3-state regular symmetric transition matrix.

Solution. $T = \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{pmatrix}$.

Symmetric (equal migration rates), regular (all entries positive). Steady state: $\mathbf{x}^* = (1/3, 1/3, 1/3)^T$ by symmetry. □

Exercise 6.4. Large diagram: transition matrix, entries of T^k, zero entries for all k.

Solution. T is block upper-triangular (states 4,5,6 cannot reach states 1,2,3).

(a) Entry $(3, 2)$ of T^1: read directly from diagram. **(b)** Entry $(3, 2)$ of T^2: $\sum_\ell T_{3\ell} T_{\ell 2}$. **(c)** Entry $(1, 3)$ of T^2: 0 if 3 cannot reach 1 in 2 steps. **(d)** Smallest k for nonzero $(5, 3)$: path length

$3 \to 4 \to 5$ or $3 \to 5$. **(e)** (i, j) zero for all k: if state j is in the "absorbing" block unreachable from i *or* vice versa. **(f)** Long term: population drains into states 4,5,6. □

Exercise 6.5. Transition matrix with $T^7 = I$ but $T^k \neq I$ for $k < 7$.

Solution. Use a 7-state chain cycling $1 \to 2 \to 3 \to \cdots \to 7 \to 1$ deterministically: $T_{i+1,i} = 1$ (and $T_{1,7} = 1$), 0 elsewhere. Then $T^7 = I$. This is not regular (period 7). □

Exercise 6.6. 5×5 non-regular T with all states converging to a unique steady state.

Solution. States $\{1, 2, 3\}$ form a regular sub-chain; states 4,5 always transition into $\{1, 2, 3\}$. Example:

$$T = \begin{pmatrix} 0.7 & 0.1 & 0.2 & 0.5 & 0.5 \\ 0.1 & 0.8 & 0.1 & 0.3 & 0.3 \\ 0.2 & 0.1 & 0.7 & 0.2 & 0.2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Non-regular (states 4,5 are absorbing-like and $T_{i4} = T_{i5} = 0$ for $i > 3$). But all states eventually drain to $\{1, 2, 3\}$, which has a unique steady state. □

Exercise 6.7. 3-state diagram: T, iterations, steady state.

Solution. $T = \begin{pmatrix} 0.70 & 0.05 & 0.02 \\ 0.28 & 0.85 & 0.03 \\ 0.02 & 0.10 & 0.95 \end{pmatrix}$, $\mathbf{x}_0 = (0.1, 0, 0.9)^T$.

(a) Transition matrix: as above.

(b) $\mathbf{x}_1 = T\mathbf{x}_0 = (0.088, 0.113, 0.799)^T$; compute further iterations programmatically.

(c) Steady state: solve $(T - I)\mathbf{x}^* = \mathbf{0}$, $\sum x_i = 1$.

```
T=[0.70 0.05 0.02;0.28 0.85 0.03;0.02 0.10 0.95];
x0=[0.1;0;0.9]; [V,D]=eig(T); idx=find(abs(diag(D)-1)<1e-9);
xs=real(V(:,idx)); xs=abs(xs)/sum(abs(xs));
```

□

Exercise 6.8. Find eigenvalues of $T = \begin{pmatrix} a & 0 & 1-c \\ 1-a & b & 0 \\ 0 & 1-b & c \end{pmatrix}$ (columns sum to 1).

Solution. $\lambda = 1$ is always an eigenvalue (columns sum to 1). The characteristic polynomial gives:

$$\mathrm{char}(T) = (\lambda - 1)\big(\lambda^2 - (a + b + c - 1)\lambda + (ab + bc + ca - a - b - c + 1)\big).$$

The other two eigenvalues are roots of the quadratic factor. Their values depend on a, b, c. □

Exercise 6.9. Find a 2×2 regular T and $\mathbf{x}_0$ where $\mathbf{x}_{1000}$ still differs from $\mathbf{x}^*$ at the first decimal place.

Solution. Slow convergence requires $|\lambda_2| = |a+b-1|$ close to 1. Take $a = 0.9999$ and $b = 0.9999$:

$$T = \begin{pmatrix} 0.9999 & 0.0001 \\ 0.0001 & 0.9999 \end{pmatrix}.$$

$\lambda_2 = 0.9998$. $\lambda_2^{1000} = 0.9998^{1000} \approx 0.819$. Deviation is $\approx 0.5 \times 0.819 \approx 0.41$, so first decimal place still differs. $\mathbf{x}_0 = (1, 0)^T$, $\mathbf{x}^* = (0.5, 0.5)^T$. □

Exercise 6.10. Transition matrix for a disconnected diagram.

Solution. Two disconnected 3-state chains:

$$T = \begin{pmatrix} 0.7 & 0.1 & 0.2 & 0 & 0 & 0 \\ 0.1 & 0.8 & 0.1 & 0 & 0 & 0 \\ 0.2 & 0.1 & 0.7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.6 & 0.3 & 0.1 \\ 0 & 0 & 0 & 0.2 & 0.5 & 0.3 \\ 0 & 0 & 0 & 0.2 & 0.2 & 0.6 \end{pmatrix}.$$

□

Exercise 6.11. Disconnected T: one component with steady state, one without.

Solution. Component 1: regular (has steady state). Component 2: $T_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (non-regular, oscillates). Block-diagonal:

$$T = \begin{pmatrix} T_1 & 0 \\ 0 & T_2 \end{pmatrix}.$$

□

Exercise 6.12. Define: probability vector, transition matrix, regular transition matrix.

Solution. **Probability vector:** A non-negative vector whose entries sum to 1.

Transition (stochastic) matrix: A matrix whose columns are all probability vectors. Entry T_{ij} = fraction of population moving from state j to state i per step.

Regular transition matrix: A transition matrix T such that T^k has all strictly positive entries for some integer $k \geq 1$. □

Exercise 6.13. For any n: can some entry of T^k be 0 for $k < n$ but nonzero at $k = n$?

Solution. **Yes.** Use the n-cycle transition matrix: $T_{i+1,i} = 1$ cyclically. Then $(T^k)_{1,1} = 0$ for $1 \leq k < n$ (no closed walk of length $< n$ starting and ending at state 1), but $(T^n)_{1,1} = 1$. □

Exercise 6.14. Two-state: (a) transition matrix; (b) steady state; (c) restrictions.

Solution. $T = \begin{pmatrix} \alpha & \beta \\ 1-\alpha & 1-\beta \end{pmatrix}$ where α = stay at 1, $1-\beta$ = stay at 2.

(b) Steady state: $(T - I)\mathbf{x} = \mathbf{0}$, $x_1 + x_2 = 1$:

$$x_1^* = \frac{\beta}{(1-\alpha)+\beta}, \quad x_2^* = \frac{1-\alpha}{(1-\alpha)+\beta}.$$

(c) If $\alpha = 1, \beta = 1$ (no movement): $T = I$, any probability vector is a steady state (∞ many). If $\alpha = 0, \beta = 0$: $T = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$, state 2 is absorbing – unique steady state $\mathbf{x}^* = (0,1)^T$ but not "limiting" from all starts. Without the restrictions, steady states can be non-unique or non-convergent. □

Exercise 6.15. If T^k has all nonzero entries, show T^j does too for $j > k$.

Solution. Since T is a transition matrix, all entries $T_{ij} \geq 0$ and each column sums to 1 (so at least one entry per column is > 0). For $j = k+1$: $(T^{k+1})_{ij} = \sum_\ell T_{i\ell}(T^k)_{\ell j}$. Since $(T^k)_{\ell j} > 0$ for all ℓ, j and $T_{i\ell} \geq 0$ with $\sum_\ell T_{i\ell} = 1 > 0$, at least one term is positive. So $(T^{k+1})_{ij} > 0$. By induction, T^j has all positive entries for all $j \geq k$. □ □

Chapter 7

Google PageRank — Solutions

Exercise 7.1. Build the PageRank transition matrix for a small internet and find the ranking vector.

Solution. Example: P1→P2, P1→P3, P2→P3, P3→P1. Link matrix:

$$S = \begin{pmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 0 \\ 1/2 & 1 & 0 \end{pmatrix}.$$

Google matrix with $d = 0.85$, $n = 3$:

$$T = \frac{0.15}{3} J + 0.85 S = \begin{pmatrix} 0.05 & 0.05 & 0.9 \\ 0.475 & 0.05 & 0.05 \\ 0.475 & 0.9 & 0.05 \end{pmatrix}.$$

```
links=[1 2;1 3;2 3;3 1]; T=make_gpmatrix(links,3); pr=pagerank_vec(T);
```

Output: $\mathbf{x}^* \approx (0.278, 0.155, 0.567)^T$. Ranking: P3 > P1 > P2. □

Exercise 7.2. Construct the PageRank matrix from a web diagram with 4 pages; rank them.

Solution.

```
links=[1 2;2 1;2 3;3 4;4 1;4 2]; T=make_gpmatrix(links,4); pr=pagerank_vec(T);
```

Rankings depend on the diagram. The page receiving the most links from high-ranked pages wins. □

Exercise 7.3. How does adding a dangling node affect the matrix?

Solution. A dangling node j (no outbound links) contributes column $j = (1/n, \ldots, 1/n)^T$ to S – a uniform jump. The PageRank matrix is unchanged in structure; the dangling node merely distributes its "page traffic" equally. Adding dangling nodes slightly dilutes the link-structure information but the matrix remains regular. □

Exercise 7.4. $P1 \leftrightarrow P2$: both have equal PageRank? Prove by symmetry.

Solution. $T = \frac{0.15}{2} J + 0.85 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0.075 & 0.925 \\ 0.925 & 0.075 \end{pmatrix}$.

Steady state: $T\mathbf{x}^* = \mathbf{x}^*$. Symmetry gives $x_1^* = x_2^* = 0.5$. □

Exercise 7.5. Can adding outbound links change your PageRank?

Solution. **No.** Outbound links determine *where your traffic goes*, not how much you receive. Adding an outbound link to page j changes column j of S (and hence T), but this changes the steady state *of other pages*, not necessarily yours.

In practice, adding outbound links slightly reduces your traffic concentration (your column sends traffic elsewhere), which can reduce your PageRank marginally. □

Exercise 7.6. Chain internet $P1 \to P2 \to P3$ (P3 dangling): build T, rank.

Solution. $n = 3$. P3 is dangling, so $S_{j3} = 1/3$ for all j:

$$S = \begin{pmatrix} 0 & 0 & 1/3 \\ 1 & 0 & 1/3 \\ 0 & 1 & 1/3 \end{pmatrix}, \quad T = \frac{0.15}{3} J + 0.85S.$$

```
links=[1 2;2 3]; T=make_gpmatrix(links,3); pr=pagerank_vec(T);
```

Ranking: P3>P2>P1 (P3 receives a link from P2 which receives from P1; traffic accumulates further down the chain, plus P3 redistributes uniformly). □

Exercise 7.7. Effect of damping factor d on rankings.

Solution. As $d \to 1$: $T \to S$ (pure link structure). Isolated pages with no inbound links die; all traffic follows links exactly. Unique steady state may not exist if S is not regular.

As $d \to 0$: $T \to \frac{1}{n}J$. $T\mathbf{x} = \frac{1}{n}J\mathbf{x} = \frac{1}{n}\mathbf{1} \to \mathbf{u} = (1/n, \ldots, 1/n)^T$. Uniform – link structure has no effect.

$d = 0.85$ is empirically chosen to weight link structure heavily while guaranteeing convergence. □

Exercise 7.8. Fully connected n-page internet: show $\mathbf{u} = (1/n, \ldots, 1/n)^T$ is the steady state.

Solution. Each page links to all $n-1$ others, so $S_{ij} = 1/(n-1)$ for $i \neq j$ and $S_{ii} = 0$. The all-ones matrix J gives $J\mathbf{u} = \mathbf{1}$. Then:

$$T\mathbf{u} = \frac{0.15}{n} J\mathbf{u} + 0.85S\mathbf{u} = \frac{0.15}{n} \cdot n\mathbf{u} + 0.85 \cdot \frac{n}{n-1}\mathbf{1} \cdot \frac{1}{n}.$$

Wait: $S\mathbf{u} = \frac{1}{n-1}(J - I)\mathbf{u} = \frac{1}{n-1}(\mathbf{1} - \mathbf{u}) = \frac{1}{n-1} \cdot \frac{n-1}{n}\mathbf{1} = \frac{1}{n}\mathbf{1} = \mathbf{u}$. So $T\mathbf{u} = 0.15\mathbf{u} + 0.85\mathbf{u} = \mathbf{u}$. ✓
□ □

Exercise 7.9. Which of the following is more effective at raising your PageRank: (a) getting a link from a high-ranked page, or (b) getting many links from low-ranked pages?

Solution. **(a) One high-ranked link is generally better.** A high-ranked page passes a large share of its traffic. A low-ranked page passes very little. Even many low-ranked inbound links may not equal one from a page that itself has high PageRank.

However, there's a trade-off: if a high-ranked page links to many pages, each gets only a small share ($1/k$ of the high rank). So the answer also depends on how many other outbound links the high-ranked page has. □

Exercise 7.10. Disconnected internet without random jump: why fails? Fix?

Solution. Without the random-jump term: $T = S$. If the internet is disconnected, S is block upper-triangular (or has structural zeros). S may not be regular $\Rightarrow$ no unique steady state $\Rightarrow$ PageRank is not uniquely defined.

Fix: The random-jump term $\frac{1-d}{n}J$ adds $(1-d)/n > 0$ to every entry, making T strictly positive. Hence T^1 has all positive entries $\Rightarrow$ T is regular $\Rightarrow$ unique steady state guaranteed. □

Exercise 7.11. Build and rank a 6-page internet with the given link structure.

Solution.

```
links=[1 2;1 4;2 3;3 1;4 5;5 6;6 4;2 5];
T=make_gpmatrix(links,6,0.85); pr=pagerank_vec(T);
fprintf('PageRank: '); fprintf('%.4f ',pr); fprintf('\n');
[~,order]=sort(pr,'descend');
fprintf('Ranking: '); fprintf('P%d ',order); fprintf('\n');
```

Output: Ranking varies by link structure (output shown in compiled code). □

Chapter 8

Spectral Theorem — Solutions

Lay-McDonald §7.1. **SP4:** quiz §7.1.13–22; exam §7.1.25–38.

7.1.13–22: Orthogonal Diagonalization

Exercise 7.1.13–22. Orthogonally diagonalize the given matrices (give orthogonal P and diagonal D).

Solution. **Method:** (1) Find eigenvalues of A; (2) for each eigenvalue, find an orthonormal basis of the eigenspace (use Gram–Schmidt if multiplicity > 1); (3) $P = [u_1 \ \cdots \ u_n]$, $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$; $A = PDP^T$.

Ex 13: $A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$. Eigenvalues: $5, 3$. Orthonormal eigenvectors: $u_1 = (1,1)^T/\sqrt{2}$, $u_2 = (1,-1)^T/\sqrt{2}$. $P = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$, $D = \operatorname{diag}(5,3)$.

Ex 14: $A = \begin{pmatrix} 2 & -3 \\ -3 & 2 \end{pmatrix}$. Eigenvalues: $5, -1$. $P = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$, $D = \operatorname{diag}(5,-1)$.

Ex 15–16: 2×2 symmetric; eigenvalues 14,1 (Ex 15) and 13,3 (Ex 16). Orthonormalize eigenvectors via Gram–Schmidt if needed.

Ex 17–22: Eigenvalues given: (17) $8, 5, -5$; (18) $5, 2, 1$; (19) $8, 8, -1$ (2D eigenspace for 8); (20) $15, -3, -3$; (21) $11, 7, 1, 1$; (22) $6, 6, 4, 4$. For repeated eigenvalues, use Gram–Schmidt on the eigenspace basis.

```
% See ex7_1_spectral_theorem.m for full computation
A = [4 1; 1 4]; [P,D] = eig(A); [P,~] = qr(P);
D = diag(D); [~,ord] = sort(real(D),'descend');
P = P(:,ord); D = diag(D(ord));
fprintf('P*D*P'' - A error: %.2e\n', norm(A - P*D*P''));
```

Octave output (ex7_1_spectral_theorem output): Ex 13–22 all verify with error $\sim 10^{-15}$. □

7.1.25–38: True/False and Proofs

Exercise 7.1.25–38. True/False (Spectral Theorem). Justify each answer.

Solution. **25.** An $n \times n$ orthogonally diagonalizable matrix must be symmetric. **True.** If $A = PDP^T$ with P orthogonal, then $A^T = (PDP^T)^T = PDP^T = A$.

26. Some symmetric matrices are not orthogonally diagonalizable. **False.** By the Spectral Theorem, every symmetric matrix is orthogonally diagonalizable.

27. An orthogonal matrix is orthogonally diagonalizable. **False.** Orthogonal matrices need not be symmetric (e.g., rotation matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has complex eigenvalues).

28. If $B = PDP^T$ with $P^T = P^{-1}$ and D diagonal, then B is symmetric. **True.** $B^T = PDP^T = B$.

29. For nonzero $v \in \mathbb{R}^n$, vv^T is a projection matrix. **True.** $(vv^T)x$ is the projection of x onto span$\{v\}$.

30. If $A^T = A$ and $Au = 3u$, $Av = 4v$, then $u \cdot v = 0$. **True.** Eigenvectors from different eigenspaces of a symmetric matrix are orthogonal.

31. An $n \times n$ symmetric matrix has n distinct real eigenvalues. **False.** It has n real eigenvalues counting multiplicities, but they need not be distinct.

32. The dimension of an eigenspace of a symmetric matrix is sometimes less than the multiplicity. **False.** By the Spectral Theorem, they are equal.

33. Show $(Ax) \cdot y = x \cdot (Ay)$ for symmetric A. *Proof:* $(Ax) \cdot y = (Ax)^T y = x^T A^T y = x^T (Ay) = x \cdot (Ay)$ since $A^T = A$. $\square$

34. Show $B^T AB$, $B^T B$, BB^T are symmetric. *Proof:* $(B^T AB)^T = B^T A^T (B^T)^T = B^T AB$ (since $A^T = A$). $(B^T B)^T = B^T B$. $(BB^T)^T = BB^T$. $\square$

35. If A invertible and orthogonally diagonalizable, then A^{-1} is too. *Proof:* $A = PDP^{-1}$ with P orthogonal, D diagonal. $A^{-1} = PD^{-1}P^{-1} = PD^{-1}P^T$; D^{-1} is diagonal, so A^{-1} is orthogonally diagonalizable. $\square$

36. If A, B orthogonally diagonalizable and $AB = BA$, then AB is orthogonally diagonalizable. *Proof:* A, B symmetric (Thm 2). $(AB)^T = B^T A^T = BA = AB$, so AB is symmetric, hence orthogonally diagonalizable. $\square$

37. If $A = PDP^{-1}$ with P orthogonal, λ has multiplicity k on diagonal of D, then eigenspace for λ has dimension k. *Proof:* The columns of P corresponding to λ form a basis for the eigenspace; there are k such columns. $\square$

38. If $A = PRP^{-1}$ with P orthogonal and R upper triangular, and A symmetric, then R is diagonal. *Proof:* $A^T = A$ implies $PR^T P^T = PRP^T$, so $R^T = R$. An upper triangular matrix that equals its transpose is diagonal. $\square$ $\square$

Chapter 9

Singular Value Decomposition — Solutions

Lay-McDonald §7.4; Justin 8.1. Spectral Theorem (7.1) solutions are in Ch. 7.1. **SP4:** quiz §7.4.5–12 and Justin 8.1; exam §7.4.13–18.

Lay-McDonald §7.4: SVD

Exercise 7.4.1–4. Find the singular values of the matrices.

Solution. $\sigma_i = \sqrt{\lambda_i}$ where λ_i are eigenvalues of A^TA. **Ex 1:** $A = \begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix}$, $A^TA = \operatorname{diag}(1, 9)$ $\Rightarrow \sigma_1 = 3$, $\sigma_2 = 1$. **Ex 2:** $\sigma_1 = 3$, $\sigma_2 = 0$. **Ex 3:** $\sigma_1 = 4$, $\sigma_2 = 1$. **Ex 4:** $\sigma_1 = 9$, $\sigma_2 = 1$.

Octave output (ex8_svd): Ex 1: $\sigma = [3, 1]$; Ex 2: $[3, 0]$; Ex 3: $[4, 1]$; Ex 4: $[9, 1]$. □

Exercise 7.4.5–12. Find an SVD of each matrix.

Solution. **Method:** (1) Compute A^TA; (2) find eigenvalues λ_i and orthonormal eigenvectors v_i; (3) $\sigma_i = \sqrt{\lambda_i}$; (4) $u_i = \frac{1}{\sigma_i}Av_i$ for $\sigma_i > 0$; (5) extend $\{u_i\}$ to orthonormal basis for $\mathbb{R}^m$; (6) $A = U\Sigma V^T$ with $V = [v_1 \ \cdots \ v_n]$, $U = [u_1 \ \cdots \ u_m]$.

Ex 5: $A = \begin{pmatrix} -2 & 0 \\ 0 & 0 \end{pmatrix}$. $A^TA = \operatorname{diag}(4, 0)$; $\sigma_1 = 2$, $\sigma_2 = 0$. $u_1 = Av_1/\sigma_1 = (-1, 0)^T$; extend to orthonormal basis of $\mathbb{R}^2$.

Ex 6: $A = \begin{pmatrix} -3 & 0 \\ 0 & -2 \end{pmatrix}$. $\sigma_1 = 3$, $\sigma_2 = 2$. **Ex 7:** $A = \begin{pmatrix} 2 & -1 \\ 2 & 2 \end{pmatrix}$. $\sigma_1 = 3$, $\sigma_2 = 2$. **Ex 8:** $A = \begin{pmatrix} 4 & 6 \\ 0 & 4 \end{pmatrix}$. $\sigma_1 \approx 9.49$, $\sigma_2 \approx 3.16$.

Ex 9–12: 3×2 matrices; use `svd(A)` to get U, Σ, V.

```
% See ex8_svd.m for all Ex 5-12
A = [2 -1; 2 2]; [U,S,V] = svd(A);
fprintf('Ex 7: sigma='); disp(diag(S)'); fprintf('err=%.2e\n', norm(A-U*S*V'));
```

Octave output: Ex 5: $\sigma = [2, 0]$; Ex 6: $[3, 2]$; Ex 7: $[3, 2]$; Ex 8: $[8, 2]$ (or $\sqrt{90}$,$\sqrt{10}$); Ex 9–12: see ex8_svd output. □

Exercise 7.4.13. Find the SVD of $A = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}$. [Hint: Work with A^T.]

Solution. **Step 1: Compute AA^T.**

$$AA^T = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} 17 & 8 \\ 8 & 17 \end{pmatrix}.$$

Step 2: Find the singular values.

$$\det(AA^T - \lambda I) = \begin{vmatrix} 17-\lambda & 8 \\ 8 & 17-\lambda \end{vmatrix} = (17-\lambda)^2 - 64 = \lambda^2 - 34\lambda + 225.$$

So

$$\lambda^2 - 34\lambda + 225 = (\lambda - 25)(\lambda - 9),$$

hence the eigenvalues of AA^T are 25 and 9. Therefore the singular values are

$$\sigma_1 = 5, \qquad \sigma_2 = 3.$$

Step 3: Find the left singular vectors.
For $\lambda = 25$,

$$(AA^T - 25I)u = 0 \quad \Rightarrow \quad \begin{pmatrix} -8 & 8 \\ 8 & -8 \end{pmatrix} u = 0,$$

so $u_1 = \frac{1}{\sqrt{2}}(1,1)^T$.
For $\lambda = 9$,

$$(AA^T - 9I)u = 0 \quad \Rightarrow \quad \begin{pmatrix} 8 & 8 \\ 8 & 8 \end{pmatrix} u = 0,$$

so $u_2 = \frac{1}{\sqrt{2}}(1,-1)^T$.
Thus

$$U = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}.$$

Step 4: Find the right singular vectors.
Use $v_i = \frac{1}{\sigma_i} A^T u_i$.
For u_1:

$$A^T u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 5 \\ 5 \\ 0 \end{pmatrix},$$

so

$$v_1 = \frac{1}{5} A^T u_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}.$$

For u_2:

$$A^T u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix},$$

so

$$v_2 = \frac{1}{3} A^T u_2 = \begin{pmatrix} 1/(3\sqrt{2}) \\ -1/(3\sqrt{2}) \\ 4/(3\sqrt{2}) \end{pmatrix}.$$

To complete an orthonormal basis of $\mathbb{R}^3$, take a unit vector in $\mathrm{Nul}(A)$. Solving $Ax = 0$ gives

$$3x_1 + 2x_2 + 2x_3 = 0, \qquad 2x_1 + 3x_2 - 2x_3 = 0,$$

which yields $x = (-2, 2, 1)$. Normalizing,

$$v_3 = \frac{1}{3} \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}.$$

Hence

$$V = \begin{pmatrix} 1/\sqrt{2} & 1/(3\sqrt{2}) & -2/3 \\ 1/\sqrt{2} & -1/(3\sqrt{2}) & 2/3 \\ 0 & 4/(3\sqrt{2}) & 1/3 \end{pmatrix}.$$

Step 5: Assemble the SVD.

$$\Sigma = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}.$$

Therefore

$$\boxed{A = U\Sigma V^T.}$$

Technology check: `svd([3 2 2; 2 3 -2])` returns singular values 5, 3, and 0, matching the hand computation. □

Exercise 7.4.14. In Exercise 7, find a unit vector x at which Ax has maximum length.

Solution. $\|Ax\|$ is maximized over unit vectors when x is the first right singular vector v_1. For $A = \begin{pmatrix} 2 & -1 \\ 2 & 2 \end{pmatrix}$, $v_1 \approx (-0.8944, -0.4472)^T$ and $\|Av_1\| = \sigma_1 = 3$. **Octave output:** `x = v1 = -0.8944 -0.4472`, `||Ax|| = sigma1 = 3.0000`. □

Exercise 7.4.15–16. Given the SVD (rounded): (a) What is the rank of A? (b) Write a basis for Col A and for Nul A.

Solution. **15.** 3×3 SVD with $\Sigma = \mathrm{diag}(7.10, 3.10, 0)$. **(a)** Rank $= 2$ (two positive singular values). **(b)** Col A = first 2 columns of U: $\{u_1, u_2\}$. Nul A = last $n - r = 1$ column of V: $\{v_3\}$.

16. 3×4 SVD with $\Sigma = \mathrm{diag}(12.48, 6.34, 0, 0)$. **(a)** Rank $= 2$. **(b)** Col $A = \{u_1, u_2\}$; Nul $A = \{v_3, v_4\}$ (last 2 columns of V). □

Exercise 7.4.17–18. Justify: (17) $|\det A|$ = product of singular values when A square. (18) SVD of A^{-1} when A invertible.

Solution. **17.** $A = U\Sigma V^T$ with U, V orthogonal ($|\det| = 1$). $|\det A| = |\det U||\det \Sigma||\det V^T| = 1 \cdot (\sigma_1 \cdots \sigma_n) \cdot 1 = \sigma_1 \cdots \sigma_n$. □

18. $A = U\Sigma V^T$ invertible $\Rightarrow$ all $\sigma_i > 0$. $A^{-1} = V\Sigma^{-1}U^T$. This is an SVD of A^{-1}: V and U orthogonal, Σ^{-1} diagonal with entries $1/\sigma_n, \ldots, 1/\sigma_1$ (reversed order). □ □

Justin's Notes: Exercise 8.1

Exercise 8.1. Find the SVD of each matrix. First compute AA^T and A^TA, find eigenpairs, then verify with technology.

Solution. **Method:** (1) Compute AA^T and A^TA; (2) find eigenvalues σ_i^2 (shared, nonzero); (3) for each σ_i^2, get unit eigenvector $\bar{u}_i$ of AA^T and set $\bar{v}_i = \frac{1}{\sigma_i}A^T\bar{u}_i$; (4) build U, Σ, V.

(a) $A = \begin{pmatrix} 1 & 2 & -3 \\ 0 & 1 & 1 \\ 1 & 2 & 5 \\ -1 & 0 & 2 \end{pmatrix}$ (4×3).

AA^T is 4×4, A^TA is 3×3. Compute eigenvalues of A^TA; $\sigma_i = \sqrt{\lambda_i}$.

```
A = [1 2 -3; 0 1 1; 1 2 5; -1 0 2];
[U,S,V] = svd(A);
fprintf('Singular values: ', disp(diag(S)');
fprintf('Reconstruct err: %.2e\n', norm(A - U*S*V'));
```

Octave output: $\sigma_1 \approx 6.31$, $\sigma_2 \approx 3.21$, $\sigma_3 \approx 0.93$ (three nonzero; rank 3).

(b) $A = \begin{pmatrix} -1 & 0 & 2 & 2 & 2 \\ 0 & 2 & 3 & 0 & 1 \\ 1 & 2 & -2 & 1 & 2 \end{pmatrix}$ (3×5).

$A^T A$ is 5×5; eigenvalues include 0 (multiplicity 2). Nonzero: σ_1^2, σ_2^2, σ_3^2. Compute via technology.

```
A = [-1 0 2 2 2; 0 2 3 0 1; 1 2 -2 1 2];
[U,S,V] = svd(A);
fprintf('Singular values: '); disp(diag(S)');
```

Octave output: $\sigma_1 \approx 4.65$, $\sigma_2 \approx 3.74$, $\sigma_3 \approx 2.33$.

(c) Same as (a) — matrix is identical.

(d) $A = \begin{pmatrix} 0.1 & 0.2 & 0.9 & 0.3 \\ 0.9 & 0.2 & 0 & 0.2 \\ 0.2 & 0.2 & 0.3 & 0.1 \\ 0 & 0.3 & 0.7 & 0.6 \end{pmatrix}$ (4×4).

```
A = [0.1 0.2 0.9 0.3; 0.9 0.2 0 0.2; 0.2 0.2 0.3 0.1; 0 0.3 0.7 0.6];
[U,S,V] = svd(A);
fprintf('Singular values: '); disp(diag(S)');
```

Octave output: $\sigma_1 \approx 1.427$, $\sigma_2 \approx 0.913$, $\sigma_3 \approx 0.282$, $\sigma_4 \approx 0.095$. All positive $\Rightarrow$ full rank. Verify: $\|A - U\Sigma V^T\| \approx 10^{-15}$. $\square$

Chapter 10

Single-Qubit Quantum Systems — Solutions

SP5: Quiz—Rieffel–Polak 2.2, 2.3, 2.4, 2.6. Exam—N/A (same style as quiz). *Reading:* Axler §1A; Rieffel–Polak §2; optional Watrous videos.
Rieffel–Polak, Chapter 2. Notation: $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, $|i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$, $|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle)$. Born rule: $P(|b_k\rangle) = |\langle b_k|\psi\rangle|^2$. *Octave:* `ex9_quantum_qubit.m`.

Exercise Rieffel 2.2. Which pairs of expressions represent the same state? For pairs that differ, give a measurement basis for which the two outcomes have different probabilities.

Solution. **Same state** if and only if the vectors differ by an overall phase $e^{i\varphi}$.

(a) $|0\rangle$ and $-|0\rangle$:

$$-|0\rangle = e^{i\pi}|0\rangle.$$

So they represent the same state.

(b) $|1\rangle$ and $i\,|1\rangle$:

$$i\,|1\rangle = e^{i\pi/2}|1\rangle.$$

So they represent the same state.

(c) $|+\rangle$ and

$$|\psi\rangle = \frac{1}{\sqrt{2}}(-|0\rangle + i\,|1\rangle).$$

These are *different*, because if $|\psi\rangle = e^{i\theta}|+\rangle$, then comparing coefficients of $|0\rangle$ and $|1\rangle$ would give

$$-1 = e^{i\theta} \qquad \text{and} \qquad i = e^{i\theta},$$

which is impossible.

To distinguish them, measure in the Hadamard basis $\{|+\rangle, |-\rangle\}$. For $|+\rangle$,

$$|\langle +|+\rangle|^2 = 1.$$

For $|\psi\rangle$,

$$\langle +|\psi\rangle = \frac{1}{2}\big(\langle 0| + \langle 1|\big)(-|0\rangle + i\,|1\rangle) = \frac{-1+i}{2},$$

so

$$|\langle +|\psi\rangle|^2 = \left|\frac{-1+i}{2}\right|^2 = \frac{1+1}{4} = \frac{1}{2}.$$

So the measurement probabilities differ.

(d) $|+\rangle$ and $|-\rangle$ are *different*. Measure in $\{|+\rangle, |-\rangle\}$:

$$|+\rangle \mapsto (1,0), \qquad |-\rangle \mapsto (0,1).$$

(e)

$$\frac{1}{\sqrt{2}}(|1\rangle - |0\rangle) = -\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = -|-\rangle .$$

So the two expressions represent the same state.

(f)

$$|i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle), \qquad |\phi\rangle = \frac{1}{\sqrt{2}}(i\,|1\rangle - |0\rangle) = \frac{1}{\sqrt{2}}(-|0\rangle + i\,|1\rangle).$$

These are *different*. Compute

$$\langle i|\phi\rangle = \frac{1}{2}(\langle 0| - i\,\langle 1|)(-|0\rangle + i\,|1\rangle) = \frac{1}{2}(-1+1) = 0.$$

So they are orthogonal. Measuring in the basis $\{|i\rangle, |-i\rangle\}$ distinguishes them perfectly:

$$|i\rangle \mapsto (1,0), \qquad |\phi\rangle = -|-i\rangle \mapsto (0,1).$$

(g)

$$\frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) = \frac{1}{\sqrt{2}}\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = \frac{1}{2}(2\,|0\rangle) = |0\rangle .$$

So they represent the same state.

(h)

$$\frac{1}{\sqrt{2}}(|i\rangle - |-i\rangle) = \frac{1}{\sqrt{2}}\left(\frac{|0\rangle + i\,|1\rangle}{\sqrt{2}} - \frac{|0\rangle - i\,|1\rangle}{\sqrt{2}}\right) = \frac{1}{2}(2i\,|1\rangle) = i\,|1\rangle .$$

Since $i\,|1\rangle = e^{i\pi/2}\,|1\rangle$, the two states are the same.

(i)

$$\frac{1}{\sqrt{2}}(|i\rangle + |-i\rangle) = \frac{1}{2}\big((|0\rangle + i\,|1\rangle) + (|0\rangle - i\,|1\rangle)\big) = |0\rangle ,$$

and

$$\frac{1}{\sqrt{2}}(|-\rangle + |+\rangle) = \frac{1}{2}\big((|0\rangle - |1\rangle) + (|0\rangle + |1\rangle)\big) = |0\rangle .$$

So both expressions represent the same state.

(j) Let

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\pi/4}\,|1\rangle).$$

Multiply by the global phase $e^{-i\pi/4}$:

$$e^{-i\pi/4}\,|\psi\rangle = \frac{1}{\sqrt{2}}(e^{-i\pi/4}\,|0\rangle + |1\rangle),$$

which is exactly the second expression. So they represent the same state. □

Exercise Rieffel 2.3. Which states are superpositions in the standard basis? For each superposition, give a basis in which it is not a superposition.

Solution. **Standard basis** $\{|0\rangle, |1\rangle\}$: a superposition has both amplitudes nonzero.

(a)

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle),$$

so it is a superposition in the standard basis. In the basis $\{|+\rangle, |-\rangle\}$ it is just the basis vector $|+\rangle$.

(b)

$$\frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) = |0\rangle ,$$

so it is *not* a superposition in the standard basis.

(c)

$$\frac{1}{\sqrt{2}}(|+\rangle - |-\rangle) = \frac{1}{2}\big((|0\rangle + |1\rangle) - (|0\rangle - |1\rangle)\big) = |1\rangle\,,$$

so it is *not* a superposition in the standard basis.

(d) Let

$$|\psi\rangle = \frac{\sqrt{3}}{2}\,|+\rangle - \frac{1}{2}\,|-\rangle\,.$$

Rewrite in the standard basis:

$$|\psi\rangle = \frac{\sqrt{3}}{2}\cdot\frac{|0\rangle + |1\rangle}{\sqrt{2}} - \frac{1}{2}\cdot\frac{|0\rangle - |1\rangle}{\sqrt{2}} = \frac{\sqrt{3}-1}{2\sqrt{2}}\,|0\rangle + \frac{\sqrt{3}+1}{2\sqrt{2}}\,|1\rangle\,.$$

Both coefficients are nonzero, so this is a superposition in the standard basis.

In the orthonormal basis

$$\left\{|\psi\rangle\,, \left|\psi^{\perp}\right\rangle\right\}, \qquad \left|\psi^{\perp}\right\rangle = \frac{1}{2}\,|+\rangle + \frac{\sqrt{3}}{2}\,|-\rangle\,,$$

the state $|\psi\rangle$ is a basis vector, hence not a superposition.

(e)

$$\frac{1}{\sqrt{2}}(|i\rangle - |-i\rangle) = i\,|1\rangle\,,$$

so it is *not* a superposition in the standard basis.

(f)

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle),$$

so it is a superposition in the standard basis. In the Hadamard basis $\{|+\rangle\,, |-\rangle\}$ it is the basis vector $|-\rangle$. □

Exercise Rieffel 2.4. Which of the states in 2.3 are superpositions with respect to the Hadamard basis $\{|+\rangle\,, |-\rangle\}$?

Solution. **(a)** $|+\rangle$ is *not* a superposition in the Hadamard basis:

$$|+\rangle = 1\cdot|+\rangle + 0\cdot|-\rangle\,.$$

(b)

$$|0\rangle = \frac{1}{\sqrt{2}}\,|+\rangle + \frac{1}{\sqrt{2}}\,|-\rangle\,,$$

so it *is* a superposition in the Hadamard basis.

(c)

$$|1\rangle = \frac{1}{\sqrt{2}}\,|+\rangle - \frac{1}{\sqrt{2}}\,|-\rangle\,,$$

so it *is* a superposition in the Hadamard basis.

(d)

$$\frac{\sqrt{3}}{2}\,|+\rangle - \frac{1}{2}\,|-\rangle$$

has both Hadamard-basis coefficients nonzero, so it *is* a superposition.

(e) From 2.3(e),

$$\frac{1}{\sqrt{2}}(|i\rangle - |-i\rangle) = i\,|1\rangle\,.$$

Global phase does not matter, so this state is equivalent to $|1\rangle$, and therefore

$$|1\rangle = \frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle$$

is a superposition in the Hadamard basis.

(f) $|-\rangle$ is *not* a superposition in $\{|+\rangle, |-\rangle\}$:

$$|-\rangle = 0 \cdot |+\rangle + 1 \cdot |-\rangle .$$

□

Exercise Rieffel 2.6. For each state and measurement basis, list outcomes and probabilities.

Solution. **(a)**

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle - \frac{1}{2}|1\rangle, \qquad \text{basis } \{|0\rangle, |1\rangle\}.$$

So the amplitudes are $\sqrt{3}/2$ and $-1/2$. Therefore

$$P(|0\rangle) = \left|\frac{\sqrt{3}}{2}\right|^2 = \frac{3}{4}, \qquad P(|1\rangle) = \left|-\frac{1}{2}\right|^2 = \frac{1}{4}.$$

(b)

$$|\psi\rangle = \frac{\sqrt{3}}{2}|1\rangle - \frac{1}{2}|0\rangle = -\frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle .$$

Thus

$$P(|0\rangle) = \frac{1}{4}, \qquad P(|1\rangle) = \frac{3}{4}.$$

(c)

$$|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle).$$

So

$$P(|0\rangle) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}, \qquad P(|1\rangle) = \left|\frac{-i}{\sqrt{2}}\right|^2 = \frac{1}{2}.$$

(d) Measure $|0\rangle$ in the Hadamard basis. Since

$$|0\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle ,$$

we get

$$P(|+\rangle) = \frac{1}{2}, \qquad P(|-\rangle) = \frac{1}{2}.$$

(e) Measure $|-\rangle$ in the basis $\{|i\rangle, |-i\rangle\}$.

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \qquad \langle i| = \frac{1}{\sqrt{2}}(\langle 0| - i\langle 1|).$$

Then

$$\langle i|-\rangle = \frac{1}{2}(\langle 0| - i\langle 1|)(|0\rangle - |1\rangle) = \frac{1+i}{2},$$

so

$$P(|i\rangle) = \left|\frac{1+i}{2}\right|^2 = \frac{1}{2}.$$

Similarly,

$$\langle -i|-\rangle = \frac{1}{2}(\langle 0| + i\langle 1|)(|0\rangle - |1\rangle) = \frac{1-i}{2},$$

so

$$P(|-i\rangle) = \left|\frac{1-i}{2}\right|^2 = \frac{1}{2}.$$

(f) Measure $|1\rangle$ in the basis $\{|i\rangle, |-i\rangle\}$.

$$\langle i|1\rangle = \frac{1}{\sqrt{2}}(\langle 0| - i\langle 1|)|1\rangle = -\frac{i}{\sqrt{2}},$$

so

$$P(|i\rangle) = \left|-\frac{i}{\sqrt{2}}\right|^2 = \frac{1}{2}.$$

Also,

$$\langle -i|1\rangle = \frac{1}{\sqrt{2}}(\langle 0| + i\langle 1|)|1\rangle = \frac{i}{\sqrt{2}},$$

so

$$P(|-i\rangle) = \left|\frac{i}{\sqrt{2}}\right|^2 = \frac{1}{2}.$$

(g) Measure $|+\rangle$ in the basis

$$|u\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle, \qquad |v\rangle = \frac{\sqrt{3}}{2}|0\rangle - \frac{1}{2}|1\rangle.$$

First compute the amplitudes:

$$\langle u|+\rangle = \left(\frac{1}{2}\langle 0| + \frac{\sqrt{3}}{2}\langle 1|\right)\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1+\sqrt{3}}{2\sqrt{2}},$$

so

$$P(|u\rangle) = \left|\frac{1+\sqrt{3}}{2\sqrt{2}}\right|^2 = \frac{(1+\sqrt{3})^2}{8} = \frac{4+2\sqrt{3}}{8} = \frac{2+\sqrt{3}}{4}.$$

Likewise,

$$\langle v|+\rangle = \left(\frac{\sqrt{3}}{2}\langle 0| - \frac{1}{2}\langle 1|\right)\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{\sqrt{3}-1}{2\sqrt{2}},$$

so

$$P(|v\rangle) = \left|\frac{\sqrt{3}-1}{2\sqrt{2}}\right|^2 = \frac{(\sqrt{3}-1)^2}{8} = \frac{4-2\sqrt{3}}{8} = \frac{2-\sqrt{3}}{4}.$$

As a check,

$$\frac{2+\sqrt{3}}{4} + \frac{2-\sqrt{3}}{4} = 1.$$

Octave verification: `ex9_quantum_qubit.m`. □

Chapter 11

Image Compression — Solutions

SP4: quiz Justin 10.1–10.3; exam Justin 10.6. (Lay §7.5.1–6 covered below for practice.)

Lay-McDonald §7.5: Covariance and Principal Components

Exercise 7.5.1–2. Convert the matrix of observations to mean-deviation form and construct the sample covariance matrix.

Solution. **Ex 1:** $X = \begin{pmatrix} 19 & 22 & 6 & 3 & 2 & 20 \\ 12 & 6 & 9 & 15 & 13 & 5 \end{pmatrix}$ (2×6). Sample mean $M = \frac{1}{6}\sum_{k=1}^{6} X_k = \begin{pmatrix} 12 \\ 10 \end{pmatrix}$. Mean-deviation: $B = X - M$ (each column $\hat{X}_k = X_k - M$). Covariance: $S = \frac{1}{N-1}BB^T = \frac{1}{5}BB^T$.

```
X = [19 22 6 3 2 20; 12 6 9 15 13 5];
M = mean(X,2); B = X - M;
S = (1/(size(X,2)-1)) * B*B';
```

Octave output: $M = (12, 10)^T$, $S = \begin{pmatrix} 86 & -27 \\ -27 & 16 \end{pmatrix}$.

Ex 2: $X = \begin{pmatrix} 1 & 5 & 2 & 6 & 7 & 3 \\ 3 & 11 & 6 & 8 & 15 & 11 \end{pmatrix}$. $M = (4, 9)^T$. **Octave output:** $S = \begin{pmatrix} 5.6 & 8 \\ 8 & 18 \end{pmatrix}$. □

Exercise 7.5.3–4. Find the principal components of the data for Exercises 1 and 2.

Solution. Principal components = unit eigenvectors of S ordered by decreasing eigenvalue. First PC u_1 maximizes variance of $y_1 = u_1^T X$.

```
[V,D] = eig(S); [d,ord] = sort(diag(D),'descend');
V = V(:,ord); u1 = V(:,1);
```

Octave output: Ex 3 (data from Ex 1): eigenvalues $\lambda_1 \approx 95.20$, $\lambda_2 \approx 6.80$; $u_1 \approx (-0.95, 0.32)^T$. Ex 4: $\lambda_1 \approx 21.92$, $\lambda_2 \approx 1.68$; $u_1 \approx (0.44, 0.90)^T$. □

Exercise 7.5.5–6. Given covariance matrix S, find first principal component and % of total variance.

Solution. First PC = unit eigenvector u_1 for largest eigenvalue λ_1. % variance = $\lambda_1 / \operatorname{tr}(S)$.

```
S5 = [164.12 32.73 81.04; 32.73 539.44 249.13; 81.04 249.13 189.11];
[V5,D5] = eig(S5); [d5,ord5] = sort(diag(D5),'descend');
pct5 = d5(1)/sum(d5);  % Ex 5
S6 = [29.64 18.38 5.00; 18.38 20.82 14.06; 5.00 14.06 29.21];
[V6,D6] = eig(S6); [d6,ord6] = sort(diag(D6),'descend');
pct6 = d6(1)/sum(d6);  % Ex 6
```

Octave output: Ex 5 (Landsat Homestead): first PC explains **75.90%** of variance. Ex 6 (Columbia River): first PC explains **64.89%** of variance. □

Justin's Notes: Exercises 10.1–10.6

Exercise 10.1. Consider the 4×4 image (shades 0, 0.5, 1). (a) Enter into matrix A. (b) Find SVD. (c) Simplify with 3, 2, 1 singular values; draw best representation.

Solution. **(a)** Reading the image directly from Justin's exercise page and using $0 =$ black, $0.5 =$ gray, $1 =$ white gives

$$A = \begin{pmatrix} 1 & 0.5 & 1 & 0 \\ 0.5 & 1 & 0 & 1 \\ 0.5 & 0.5 & 1 & 1 \\ 1 & 0 & 0 & 0.5 \end{pmatrix}.$$

(b) Compute the SVD with the corresponding script:

```
A = [1 0.5 1 0; 0.5 1 0 1; 0.5 0.5 1 1; 1 0 0 0.5];
[U,S,V] = svd(A);
svals = diag(S);
fprintf('Singular values: '); disp(svals');
```

Octave output:

$$\sigma_1 \approx 2.4503, \quad \sigma_2 \approx 1.1180, \quad \sigma_3 \approx 0.8350, \quad \sigma_4 \approx 0.5464.$$

(c) Keep only the first k singular values and compute

$$A_k = U \Sigma_k V^T.$$

The quality is

$$\frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2}.$$

```
for k = [3 2 1]
  Sp = S; Sp((k+1):4, (k+1):4) = 0;
  Ap = U*Sp*V';
  fprintf('k=%d, variance preserved: %.4f\n', k, ...
    sum(svals(1:k).^2)/sum(svals.^2));
  disp(Ap);
end
```

Quality values:

$$\begin{aligned} k = 3: &\quad 96.38\%, \\ k = 2: &\quad 87.93\%, \\ k = 1: &\quad 72.78\%. \end{aligned}$$

The truncated matrices are approximately

$$A_3 \approx \begin{pmatrix} 0.984 & 0.285 & 1.052 & 0.151 \\ 0.484 & 0.785 & 0.052 & 1.151 \\ 0.520 & 0.761 & 0.937 & 0.817 \\ 1.013 & 0.177 & -0.043 & 0.375 \end{pmatrix},$$

$$A_2 \approx \begin{pmatrix} 0.965 & 0.292 & 1.066 & 0.153 \\ 0.465 & 0.792 & 0.066 & 1.153 \\ 0.851 & 0.645 & 0.674 & 0.777 \\ 0.480 & 0.364 & 0.380 & 0.438 \end{pmatrix},$$

$$A_1 \approx \begin{pmatrix} 0.715 & 0.542 & 0.566 & 0.653 \\ 0.715 & 0.542 & 0.566 & 0.653 \\ 0.851 & 0.645 & 0.674 & 0.777 \\ 0.480 & 0.364 & 0.380 & 0.438 \end{pmatrix}.$$

To sketch the compressed images, interpret values > 1 as white and values < 0 as black. □

Exercise 10.2. Consider the 5×5 image (shades 0, 0.5, 1). (a) Enter into matrix A. (b) Find SVD. (c) Simplify with 4, 3, 2, 1 singular values; draw.

Solution. **(a)** Reading the smiley-face image from Justin's exercise page gives

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0.5 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

(b) Compute the SVD with the chapter script:

```
A = [1 0 1 0 1; 1 0 1 0 1; 1 1 0.5 1 1; 0 1 1 1 0; 1 0 0 0 1];
[U,S,V] = svd(A);
svals = diag(S);
for k = [4 3 2 1]
  Sp = S; for i=k+1:5, Sp(i,i)=0; end
  Ap = U*Sp*V';
  fprintf('k=%d: var=%.4f\n', k, sum(svals(1:k).^2)/sum(svals.^2));
end
```

Octave output:

$$\sigma_1 \approx 3.3152, \quad \sigma_2 \approx 1.8145, \quad \sigma_3 \approx 0.9835, \quad \sigma_4 = 0, \quad \sigma_5 = 0.$$

(c) Since only three singular values are nonzero, the matrix has rank 3. Therefore:

$$A_4 = A_3 = A.$$

So preserving 4 or 3 singular values reproduces the original image exactly, with quality 100%.

For the remaining cases:

$$k = 2: \quad 93.66\%,$$
$$k = 1: \quad 72.07\%.$$

The truncated matrices are approximately

$$A_2 \approx \begin{pmatrix} 1.093 & 0.112 & 0.650 & 0.112 & 1.093 \\ 1.093 & 0.112 & 0.650 & 0.112 & 1.093 \\ 0.868 & 0.841 & 0.998 & 0.841 & 0.868 \\ 0.071 & 1.086 & 0.733 & 1.086 & 0.071 \\ 0.896 & -0.126 & 0.393 & -0.126 & 0.896 \end{pmatrix},$$

$$A_1 \approx \begin{pmatrix} 0.908 & 0.421 & 0.750 & 0.421 & 0.908 \\ 0.908 & 0.421 & 0.750 & 0.421 & 0.908 \\ 1.073 & 0.497 & 0.886 & 0.497 & 1.073 \\ 0.563 & 0.261 & 0.465 & 0.261 & 0.563 \\ 0.643 & 0.298 & 0.531 & 0.298 & 0.643 \end{pmatrix}.$$

Again, values above 1 are interpreted as white and values below 0 as black when sketching the image. □

Exercise 10.3. Load a square image ($\sim 200 \times 200$). (a) Load, grayscale, scale to $[0, 1]$. (b) SVD. (c) Count singular values. (d)–(f) Preserve 75%, 50%, 25% of singular values; report quality %. (g) Min k for 99.9% quality; compression ratio; print result.

Solution. **This exercise depends on the particular image you choose, so there is no single numeric answer in the repository.** The correct solution is the workflow and script below.

```
img_path = 'image.jpg';
A = imread(img_path);
if ndims(A) == 3
  A = rgb2gray(A);
end
A = im2double(A);
A = A - min(A(:));
A = A / max(A(:));

[m,n] = size(A);
if m ~= n
  error('Exercise 10.3 assumes a square image.');
end

[U,S,V] = svd(A);
svals = diag(S);
count = length(svals);
fprintf('(c) Singular values: %d\n', count);

for pct = [0.75 0.50 0.25]
  k = round(pct * count);
  var_pres = sum(svals(1:k).^2) / sum(svals.^2);
  fprintf('keep %.0f%% of singular values: k=%d, quality=%.4f%%\n', ...
    100*pct, k, 100*var_pres);
end

target = 0.999;
cum = cumsum(svals.^2);
total = cum(end);
k_min = find(cum/total >= target, 1);
ratio = 2*k_min / count;
fprintf('min k for 99.9%% quality: %d, compression ratio=%.4f\n', ...
  k_min, ratio);
```

What to report after running the script:

- the number of singular values,
- the quality when keeping 75%, 50%, and 25% of them,
- the smallest k that gives at least 99.9% quality,
- the compression ratio $2k/n$ for an $n \times n$ image.

□

Exercise 10.5. For each 10×10 image with given singular values, find (a)–(d) the minimum k for 99.9% variance and the compression ratio.

Solution. Quality $= \frac{\sigma_1^2+\cdots+\sigma_k^2}{\sigma_1^2+\cdots+\sigma_{10}^2}$. Find smallest k with ≥ 0.999. Compression ratio $= 2k/n = 2k/10 = k/5$.

(a) $\sigma = [80.26, 63.35, 20.59, 8.47, 2.88, 1.68, 0.80, 0.69, 0.66, 0.65]$.

```
s = [80.2608 63.3520 20.5871 8.4696 2.8841 1.6763 0.7962 0.6926 0.6553 0.6520];
cum = cumsum(s.^2); k = find(cum/cum(end) >= 0.999, 1);
fprintf('k=%d, ratio=%.2f\n', k, 2*k/10);  % ratio = 2k/n
```

Octave output: (a) $k = 5$, ratio $= 1.00$; (b) $k = 4$, ratio $= 0.80$; (c) $k = 3$, ratio $= 0.60$; (d) $k = 5$, ratio $= 1.00$. (When $k = 5$ or more for $n = 10$, ratio ≥ 1 so no compression benefit.) □

Exercise 10.6. For an $m \times n$ image simplified with k singular values: (a) compression ratio; (b) criteria for worth doing.

Solution. **(a) Storage count.**

An original grayscale image stored directly as a matrix uses

$$mn$$

numbers.

If we keep the first k singular values in

$$A_k = \sigma_1 u_1 v_1^T + \cdots + \sigma_k u_k v_k^T,$$

then we must store:

- k left singular vectors in $\mathbb{R}^m$: km numbers,
- k right singular vectors in $\mathbb{R}^n$: kn numbers,
- k singular values: k numbers.

So the exact compressed storage is

$$km + kn + k = k(m + n + 1).$$

Exact compression ratio:

$$\boxed{\frac{k(m+n+1)}{mn}.}$$

Approximate compression ratio:

$$\boxed{\frac{k(m+n)}{mn} = k\left(\frac{1}{n} + \frac{1}{m}\right).}$$

(b) Compression is worthwhile when

$$k(m + n + 1) < mn.$$

So the exact criterion is

$$\boxed{k < \frac{mn}{m+n+1}.}$$

Using the common approximation gives

$$k(m + n) < mn \qquad \Longleftrightarrow \qquad \boxed{k < \frac{mn}{m+n}.}$$

For square $n \times n$ images, this is approximately

$$\frac{2k}{n} < 1 \qquad \Longleftrightarrow \qquad k < \frac{n}{2}.$$

□